Prepared in cooperation with Metropolitan Water District and State Water Contractors

Environmental Characteristics of Select Managed Ponds in the Sacramento–San Joaquin Delta: Implications for Native Fish Conservation and Research

Open-File Report 2025–1040

U.S. Department of the Interior
U.S. Geological Survey

Cover. Managed pond located on Bacon Island, San Joaquin County, California. Photograph by Frederick Feyrer, U.S. Geological Survey, December 20, 2021.

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Conversion Factors

International System of Units to U.S. customary units

Multiply	By	To obtain
	Length	
meter (m)	3.281	foot (ft)
	Mass	
gram (g)	0.03527	ounce, avoirdupois (oz)
	Area	
hectare (ha)	2.47105	acre

Temperature in degrees Celsius (°C) may be converted to degrees Fahrenheit (°F) as follows:

$$°F = (1.8 \times °C) + 32.$$

Datum

Horizontal coordinate information is referenced to the North American Datum of 1983 (NAD 83).

Supplemental Information

Specific conductance is in microsiemens per centimeter at 25 degrees Celsius (µS/cm at 25 °C).

Concentrations of chemical constituents in water are in either milligrams per liter (mg/L) or micrograms per liter (µg/L).

Abbreviations

DOC	dissolved organic carbon
ELISA	enzyme-linked immunosorbent assay
EPA	U.S. Environmental Protection Agency
FNU	Formazin Nephelometric Units
GC	gas chromatography
LC	liquid chromatography
MS/MS	tandem mass spectrometry
PCA	principal components analysis
PSU	practical salinity units
USGS	U.S. Geological Survey

Environmental Characteristics of Select Managed Ponds in the Sacramento–San Joaquin Delta: Implications for Native Fish Conservation and Research

By Frederick V. Feyrer,[1] Shawn Acuña,[2] Jordan M. Buxton,[1] Ethan R. Enos,[1] Michelle L. Hladik,[1] James Orlando,[1] and Matthew J. Young[1]

Abstract

The use of wetlands to support native fish research and conservation efforts in the Sacramento–San Joaquin Delta (Delta) of California is a growing priority. The purpose of our study was to examine the physiochemical and biological characteristics of select managed ponds in the Delta to determine if they would be suitable habitats for research involving the conservation of delta smelt (*Hypomesus transpacificus*). We studied 10 managed ponds distributed across the central part of the Delta situated on Bacon Island and Bouldin Island in San Joaquin County, and Holland Tract and Webb Tract islands in Contra Costa County. The managed ponds had a diversity of physical habitat configurations and were not directly connected to waterways surrounding the islands and, therefore, not affected by tides. We studied the managed ponds from approximately November 2021 to December 2023 to assess water quality, zooplankton, fish, and pesticide metrics. Water levels in the managed ponds were managed to varying degrees and were mostly independent of climate-driven wet-dry seasonality. Water quality conditions varied among ponds and were independent of geographic location. Overall, mean monthly chlorophyll *a* concentration ranged from 15 to 57 (mean=30) micrograms per liter (µg/L), dissolved oxygen concentration ranged from 4 to 9 (mean=7) milligrams per liter (mg/L), pH was 8, salinity was 1 practical salinity units (PSU), specific conductance ranged from 1,202 to 1,839 (mean=1,471) microsiemens per centimeter (µS/cm), and turbidity ranged from 13 to 24 (mean=19) Formazin Nephelometric Units (FNU). Water temperature thresholds that contribute to stress (21 degrees Celsius [°C]) and mortality (28 °C) of delta smelt were often exceeded during summer and fall, though vertical stratification contributed to lower bottom temperatures in the deepest managed ponds, which could potentially provide thermal refugia for delta smelt so long as dissolved oxygen concentrations are suitable. Zooplankton populations were broadly similar among managed ponds and included calanoid and cyclopoid copepods that would be suitable prey for delta smelt. Overall average total zooplankton biomass, as measured with a Schindler-Patalas trap, was 0.6 µg/L (min=0, max=63.6) and peaked during spring at more than 4 µg/L. Fish populations highly varied among the managed ponds with potential predators of delta smelt such as largemouth bass (*Micropterus salmoides*) and black crappie (*Pomoxis nigromaculatus*) present in several of the managed ponds; predator distribution among ponds seemed to have been driven primarily by deliberate stocking to facilitate local fisheries. Measured pesticide concentrations were below U.S. Environmental Protection Agency Aquatic Life Benchmarks except for exceedances of three compounds (diuron [herbicide], clothianidin [insecticide], and deltamethrin [pyrethroid insecticide]) in samples collected from ponds on Bouldin Island and Webb Tract. Overall, most managed ponds seemed suitable to support delta smelt, though physical control of potential predators and summer temperature might be needed. The results provide guidance on how to engineer and manage new managed ponds to support research and conservation efforts for delta smelt and other native fishes.

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Introduction

The Sacramento–San Joaquin Delta (Delta) of California was historically a vast inland wetland system (Whipple and others, 2012). Land surface elevation of wetlands kept pace with sea level rise and produced organic peat soils up to 20 meters (m) in depth (Drexler, 2011). The natural system was heavily disturbed and altered starting in the late 1800s when about 95 percent of Delta wetlands were diked and drained to create “islands” for agricultural use (Drexler and others, 2009; Whipple and others, 2012). The drainage of wetlands and associated agricultural practices have contributed to the loss of peat soils and resulted in land surface subsidence up to 7 m below sea level on Delta islands (Weir, 1950; Prokopovich, 1985; Deverel and Rojstaczer, 1996; Lund and others, 2007). This habitat alteration and land subsidence has greatly affected the Delta’s ecology, increased carbon emissions, and, at present, pose a substantial threat to the integrity of the levees built to create the islands (Lund and others, 2007).

Restoration has been conceived as a tool to mitigate the loss of historic wetlands. Restoration in the form of creating managed impounded wetlands on Delta islands has the potential to halt oxidative loss of peat soils and may accrete organic matter, thereby gaining land surface elevation (Miller and others, 2008; Miller and Fujii, 2011). Restored wetlands also may play an important but complicated role for mitigating the effects of climate change. Converting drained peat soils to wetlands can sequester carbon but also may generate high methane emissions (Hemes and others, 2019; Arias-Ortiz and others, 2021). Results of studies of restored experimental wetlands on Delta islands indicate climate benefits are highly variable and may take upwards of two to eight decades to generate net positive greenhouse gas benefits (Deverel and others, 2014; Chamberlain and others, 2018). Considering slow and uncertain land surface elevation gains and climate benefits, there is interest in identifying additional beneficial uses of restored wetlands on Delta islands for conservation purposes. One such purpose under consideration is to use managed wetland ponds as tools to support native fish conservation efforts. A fundamental first step for this effort is examining managed pond habitat characteristics to assess their suitability to support native fish.

The purpose of our study was to examine the physiochemical and biological characteristics of select managed ponds on Delta islands to determine if they would be suitable habitats for studies focused on the conservation of delta smelt (*Hypomesus transpacificus*). Delta smelt is an imperiled species thought to be on the brink of extinction (Moyle and others, 2016; Hobbs and others, 2017). Recovery of delta smelt is important, among many reasons, because its critical habitat, the Delta, is a key component of California’s water supply and is subject to management actions to protect delta smelt that may affect water supply (Moyle and others, 2018). Supplementation of the wild delta smelt population with artificially propagated fish is considered a vital step in preventing extirpation from the wild (Lessard and others, 2018; Hung and others, 2019). Opportunities for research and rearing of artificially propagated delta smelt are currently limited (Lindberg and others, 2013). Our objective was to determine if managed ponds on Delta islands could be useful tools for research and development aspects of delta smelt reintroduction. One potential application would be to expand the limited existing infrastructure by using managed ponds to rear artificially propagated delta smelt before their release into Delta sloughs or channels (Maynard and others, 2004; Garlock and others, 2014).

Study Area

Our study involved a total of 10 managed ponds distributed across four separate central Delta islands: Bacon Island, Bouldin Island, Holland Tract, and Webb Tract (fig. 1). The managed ponds do not have official names, so we refer to them in a geographical context (Bacon North, and so on; table 1; fig. 1). The managed ponds have a variety of origins, including (1) borrow pits that were created from excavation of material used to repair levees (Bouldin East and Bouldin West), (2) eroded depressions that were not reclaimed after island flooding caused by levee breaks (Holland North), and (3) habitats created and managed for a variety of recreation and conservation purposes (Bacon North, Bacon South, Holland Middle, Holland South, Webb North, Webb Middle, and Webb East). The managed ponds had a diversity of physical configurations (table 1). They ranged in perimeter length from 470 m (Holland Middle) to 2,089 m (Webb East), ranged in surface area from 0.9 hectares (ha; Holland Middle) to 9.1 ha (Bacon North), and ranged in maximum depth from 0.7 m (Holland Middle) to 5.3 m (Bouldin East; table 1). Agriculture was the dominant land use on all islands, and pond perimeters were variably (0 to 100 percent) buffered from agricultural activities by riparian habitat that was broadly characterized into two groups: (1) tules, including other emergent wetland vegetation; and (2) forest, including cottonwoods, willows, and other woody plants (table 1). The managed ponds were not directly connected to sloughs surrounding the islands and were, therefore, not affected by tides. Water levels in the managed ponds were managed to varying degrees for different purposes based on land use activities on specific islands. Water is transported on and off the islands from adjacent sloughs via pumps and can be moved within islands through small canals and other infrastructure; water management activities are not documented quantitatively, and therefore, specific corresponding data are not available. Thus, water depth and associated volume was variable and largely independent of climatic wet-dry seasonality (fig. 2). The Holland Middle and South ponds, which lack inlet/outlet canals, dried up during this study.

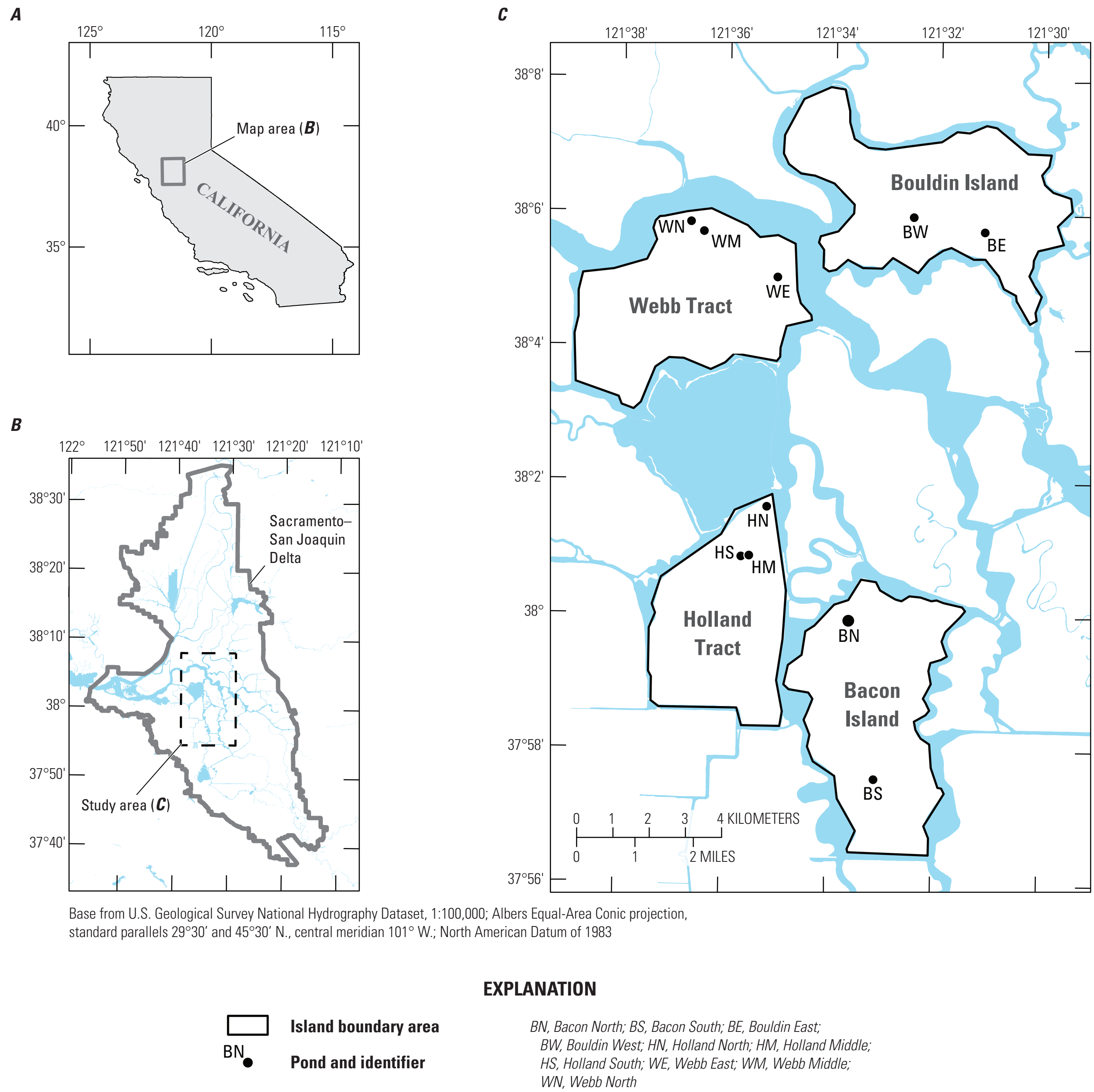

Figure 1. *A*, Location of the Sacramento–San Joaquin Delta in California; *B*, study area within the Sacramento–San Joaquin Delta; and *C*, managed pond study sites.

Table 1. Managed pond study site physical characteristics within the Sacramento–San Joaquin Delta in California.

[ha, hectare; m, meter; %, percentage]

Managed pond	Perimeter (m)	Surface area (ha)	Depth (m)	Riparian habitat composition (%)		
				Agriculture	Tules	Forest
Bacon North	1,692	9.1	4.2	59	0	41
Bacon South	852	2.9	5.1	37	63	0
Bouldin East	881	4.2	5.3	22	67	11
Bouldin West	1,336	9.6	3.6	61	39	0
Holland Middle	470	0.9	0.7	0	69	31
Holland North	1,449	4.1	3.7	0	27	73
Holland South	1,055	2.5	2.4	0	38	62
Webb East	2,089	1.9	1.8	3	56	41
Webb Middle	850	2.6	0.8	27	55	18
Webb North	1,214	6.3	1.1	15	59	26

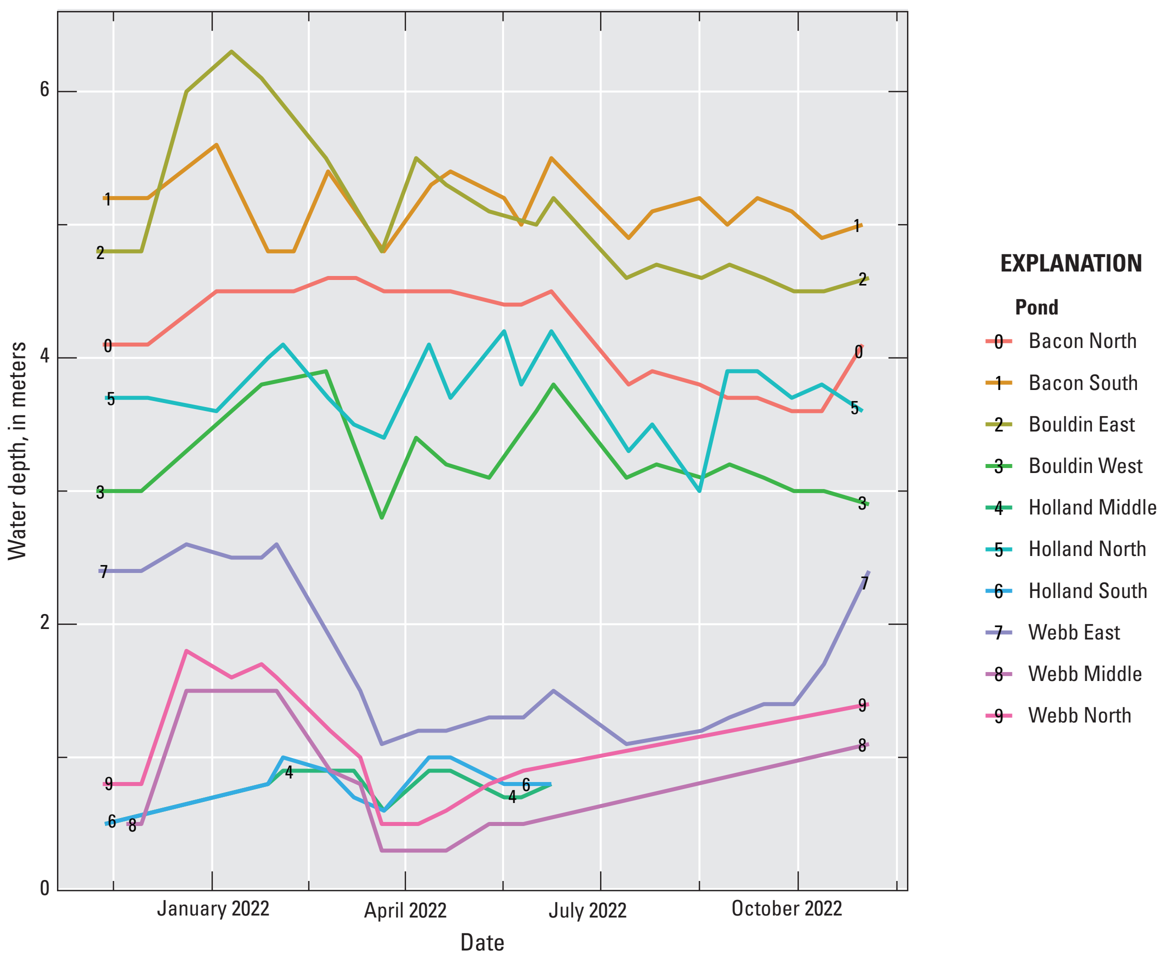

Figure 2. Water depth time series for managed pond study sites within the Sacramento–San Joaquin Delta in California. The Holland Middle and Holland South Pond study sites do not have complete time series because they dried up during the study.

Methods

We sought to generate a comprehensive baseline assessment of the overall physiochemical and biological characteristics of each managed pond. Our approach involved characterizing aspects of (1) water quality parameters using a combination of continuous and discrete measurements; (2) pesticide concentrations (the primary contaminants of concern) in water, suspended sediment, and bed sediment of managed ponds and adjoining inlet/outlet canals; (3) zooplankton species composition and density; and (4) fish occupancy. The study period was from November 2021 to December 2022.

Water Quality

Various water quality parameters were measured during the duration of the full study period. This included continuous (every 15 minutes) measurements of water temperature (in degrees Celsius [°C]) recorded by HOBO Pro v2 data loggers (Onset, Bourne, Massachusetts). Data loggers were positioned at the approximate center of each managed pond, near the surface and bed of managed ponds greater than 1.5 m in depth (Bacon North, Bacon South, Bouldin East, Bouldin West, Holland North, and Webb East) and only at mid-depth (0.75 m) in managed ponds less than 1.5 m in depth (app. 1; table 1). Discrete measurements of water temperature (°C), dissolved oxygen concentration in milligrams per liter (mg/L), specific conductance in microsiemens per centimeter (µS/cm), salinity in practical salinity units (PSU), turbidity in Formazin Nephelometric Units (FNU), chlorophyll *a* concentration in micrograms per liter (µg/L), and pH were obtained from handheld YSI EXO2 sondes (Yellow Springs Instruments, Yellow Springs, Ohio). Discrete measurements were made approximately every 2 weeks at the same locations within ponds where the HOBO loggers were positioned.

Additional water quality parameters were measured seasonally, once during winter and once during summer, to characterize baseline conditions. In these seasonal events, we measured concentrations of nutrients (ammonia, nitrate, nitrite, total nitrogen, organic nitrogen, and orthophosphate), metals (barium, calcium, magnesium, manganese, selenium, strontium, and zinc), sediment (suspended), dissolved organic carbon (DOC), chlorophyll *a* and pheophytin *a*, and inorganic aspects of water quality (silica and hardness). Water samples for analysis were collected with a Van Dorn sampler at the same locations and depths within ponds where the HOBO loggers were positioned. Water samples for suspended sediment analysis were collected in 1-liter (L) plastic bottles, stored at room temperature, and analyzed at the U.S. Geological Survey (USGS) Sediment Lab in Santa Cruz, Calif. Water samples collected for chlorophyll *a*, metals, and nutrients were stored on wet ice (less than 24 hours) and refrigerated until filtered in the laboratory. Chlorophyll *a* samples were prepared by filtering sample water over a 47 millimeter (mm) diameter, 1.7-micrometer (µm) pore size, pre-combusted glass fiber filter. Water samples for metals, nutrients, and dissolved organic carbon analyses were filtered through a 0.45-µm Pall capsule filter using a peristaltic pump. Water samples for metal analysis were collected in a 250-milliliter (mL) acid-rinsed clear polyethylene bottle and preserved with 2 mL of 7.5 normal (N) nitric acid (Ultrex HNO_3). Water samples for nutrient analysis were collected in 125-mL amber-plastic bottles. Water samples for DOC analysis were collected in 125-mL glass-amber bottles and preserved with 1 mL of 4.5 N sulfuric acid (H_2SO_4). Chlorophyll *a*, metals, nutrients, and DOC samples were analyzed by the USGS National Water Quality Laboratory in Denver, Colorado.

Pesticides

Pesticides were assessed at three separate sampling events, which we term surveys, occurring in March 2022, June 2022, and February 2023. Survey periods were chosen to represent wet and dry seasons. Water, suspended sediment, and bed sediment samples were collected in each managed pond. Additional water samples were collected from select adjoining inlet/outlet canals (app. 2). Managed pond water samples for pesticide analysis were collected in 1-L amber-glass bottles using a weighted bottle sampler at approximately 0.5 m depth. Managed pond water samples for glyphosate analysis were collected by submerging a 50-ml amber-glass vial to a depth of 0.5 m. Managed pond bed sediment samples were collected using an Ekman dredge. The top 2 centimeters (cm) of depositional sediment collected within the dredge was subsampled using a stainless-steel scoop and placed in 250-ml amber-glass jars. Managed pond inlet water samples were collected by hand submerging 1-L and 50-ml amber-glass bottles below the water surface. Managed pond inlet bed sediment samples were collected using a stainless-steel scoop to transfer the top 2 cm of depositional sediment from multiple sites within 1–3 m of each other into 250-ml amber-glass jars.

All samples were placed on wet ice immediately after collection and delivered to the USGS Organic Chemistry Research Laboratory in Sacramento, Calif., for processing and analysis. Water samples and associated filtered residues (suspended sediment) were processed following established procedures (Gross and others, 2024). Bed sediment samples were processed following Black and others (2023).

Water, suspended sediment, and bed sediment samples were analyzed using liquid chromatography (LC) and gas chromatography (GC) with tandem mass spectrometry (MS/MS) following procedures described in Gross and others (2024). Water samples were analyzed for 178 pesticides, suspended sediment samples were analyzed for 173 pesticides, and bed sediment samples were analyzed for 162 pesticides (app. 3, table 3.1). Samples were analyzed using LC/MS/MS followed by GC/MS/MS. Glyphosate samples were analyzed using an enzyme-linked immunosorbent assay (ELISA) microtiter plate (Gold Standard Diagnostics, Davis, California). Analysis was completed on a ChroMate microplate reader (RayBiotech, Peachtree Corners, Georgia) and the absorbance was read at 450 nanometers (nm). Evaluation was performed using a 4-parameter logistic regression. All samples were analyzed in duplicate, and the reported concentrations were the average of the duplicate readings. Method detection and reporting limits for water and suspended sediment pesticide analyses were determined by U.S. Environmental Protection Agency (EPA) guidelines (U.S. Environmental Protection Agency, 2016) and described in Gross and others (2024). For glyphosate analyses by ELISA, the least detectable dose was 50 nanograms per liter (ng/L) and concentrations less than 75 ng/L are below the reporting limit.

Zooplankton

Zooplankton species composition and density were assessed approximately once per month. Samples to characterize zooplankton were collected using a 12-L Schindler-Patalas plankton trap fitted with a 200-ml dolphin bucket with 61-micron mesh and Nitrex filter net (54-mm cod end, 311-mm long, 63 micron). The Schindler-Patalas trap was used because variable water depth and vegetation density within and among managed ponds made it impossible to collect consistent, standardized samples with other methods, such as nets towed or retrieved through the water column. One sample was collected at a depth of approximately 1 m at the same geographical locations within managed ponds where the HOBO loggers were positioned. Samples were preserved in the field in 10-percent formalin and analyzed by a contractor (EcoAnalysts, Moscow, Idaho).

Fish

Each managed pond was comprehensively surveyed once to determine the presence-absence of individual fish species. One or more sampling gear types and methods—beach seine, gillnet, and boat electrofishing—were used, as feasible, in individual managed ponds based on their physical habitat in attempt to capture all species present. Beach seining was done in Bacon North, Bouldin East, Bouldin West, Holland Middle, and Holland South. Electrofishing was done in Bacon South, Bouldin East, Bouldin West, Holland North, Webb East, Webb Middle, and Webb North. Gillnetting was done in Bouldin East, Bouldin West, Holland North, and Webb East. Sampling was done in March and April 2022. The beach seine measured 6 × 1.2 m with 3-mm mesh. The gillnet measured 45.7 × 1.8 m with five equal length panels of 38-, 51-, 64-, 76-, and 89-mm mesh. Electrofishing was done with a Smith Root Model Generator Powered Pulsator pulsed direct current unit powered by a 5.5-horsepower generator mounted on a 3.6-m aluminum boat. All captured fishes were identified to species and released alive.

Data Analysis

Tabular and graphical summaries of data were constructed to assess patterns of measured parameters within and among ponds during the study period. For example, continuous water temperature data were plotted in time series with reference to temperature threshold values that contribute to stress (21 °C) and mortality (28 °C) of delta smelt (Hung and others, 2022). In two cases, we ran a principal components analysis (PCA) to statistically characterize dominant modes of variability within and among managed pond specific characteristics (R Core Team, 2020). These two cases involved: (1) discrete water quality measurements and (2) zooplankton species composition data. For the water quality PCA, we examined select water quality variables measured each month (temperature, dissolved oxygen concentration, turbidity, specific conductance, chlorophyll concentration, and pH), and included measurements taken at the surface or middle depth of each managed pond. For the zooplankton PCA, we included adult life stages of taxa, which consisted of greater than or equal to 1 percent of the overall total biomass reported across all managed ponds. It is noted that we also ran extensive exploratory modeling of data to identify relationships among specific managed ponds, water quality, zooplankton, and fish but ultimately deemed such analyses uninformative because it was apparent that pond management activities (for example, water level management and fish introductions) overwhelmed natural ecological processes.

Data Availability

Continuous and discrete water quality, zooplankton, and fish data are available in a USGS data release from Buxton and others (2023). Seasonal water quality and pesticide data are available from U.S. Geological Survey (2024). Full site lists are in appendixes 1 and 2.

Results

Water Quality

Continuous water temperature measurements indicated that water temperatures ranged from approximately 8 °C in winter to approximately 30 °C in summer, with patterns that were generally similar among ponds (fig. 3), except for Bacon South at the bottom. Seasonal vertical stratification of temperature was also observed. Bottom temperatures were in general, approximately 2–5 °C cooler than surface temperatures from approximately May to October (fig. 3). The delta smelt temperature stress threshold value of 21 °C was exceeded from May through October at the bottom and surface, with occasional surface exceedances also occurring as early as March and as late as November (table 2; fig. 3). The delta smelt temperature mortality threshold value of 28 °C was exceeded intermittently only at the surface from May through October (table 2; fig. 3).

Monthly discrete water quality measurements indicated a high degree of variability within and among ponds for all measured parameters (fig. 4). Overall, grand mean monthly (averaged across all ponds) chlorophyll *a* concentration was 30 µg/L, dissolved oxygen concentration was 7 mg/L, pH was 8, salinity was 1 PSU, specific conductance was 1,471 µS/cm, and turbidity was 19 FNU. Ponds greater than 1.5 m in depth (those in which we had surface and bottom measurements) indicated varying degrees of vertical stratification across all parameters as measured by percent difference of surface-to-bottom values (fig. 5). The most consistent pattern of vertical stratification was with temperature, dissolved oxygen concentration, and pH values being generally higher at surface than at bottom from approximately spring to summer, particularly in the deepest ponds (Bacon South, Bacon North, Holland North, Bouldin East, and Bouldin West; fig. 5). Results of the water quality PCA indicated water quality conditions were independent of geography with varying degrees of similarity among individual managed ponds (table 3; fig. 6). On Bacon Island, Bacon North was generally more turbid with higher pH and specific conductance than Bacon South. Similarly, on Bouldin Island, Bouldin West was generally more turbid with higher pH and specific conductance than Bouldin East. Holland Tract managed ponds indicated generally similar water quality conditions except that Holland Middle and Holland South had elevated salinity and specific conductance. Webb Tract managed ponds indicated the highest similarity in water quality parameters among managed ponds on an individual island.

Seasonal water quality measurements indicated high variability in the measured parameters among managed ponds and islands (table 4). See table 4 for absolute values of all measured parameters. Although overall variability was high, within islands, the measured parameters indicated the highest similarity among ponds on Webb Tract. Holland Middle and Holland South generally indicated the highest values across all non-organic parameters. Holland North indicated the highest concentrations of nutrient parameters (total nitrogen=3.1 mg/L; orthophosphate=0.975 mg/L; ammonia=1.956 mg/L). Among the organic parameters, DOC was highest in Bouldin West (58 mg/L) and chlorophyll *a* was highest in Bacon North (138.2 µg/L).

Figure 3. Maximum daily water temperature time series for surface and bottom of managed ponds greater than 1.5 meters (m) in depth within the Sacramento–San Joaquin Delta in California. Temperature ranges exceeding 21 degrees Celsius (°C) and 28 °C are indicated with colored shading and may potentially contribute to delta smelt stress and mortality, respectively (Hung and others, 2022). Data summarized from Buxton and others (2023).

Table 2. Percentage of days that the daily maximum water temperature exceeded delta smelt temperature stress threshold of 21 degrees Celsius (°C) and temperature mortality threshold of 28 °C in managed ponds within the Sacramento–San Joaquin Delta in California. Data summarized from Buxton and others (2023).

[≥, greater than or equal to]

Pond	Position	Percentage of days ≥21 °C delta smelt stress threshold	Percentage of days ≥28 °C delta smelt mortality threshold
Bacon North	Surface	46	6
	Bottom	38	0
Bacon South	Surface	46	4
	Bottom	6	0
Bouldin East	Surface	44	9
	Bottom	30	0
Bouldin West	Surface	41	0
	Bottom	35	0
Holland North	Surface	48	7
	Bottom	32	0
Webb East	Surface	47	9
	Bottom	38	0

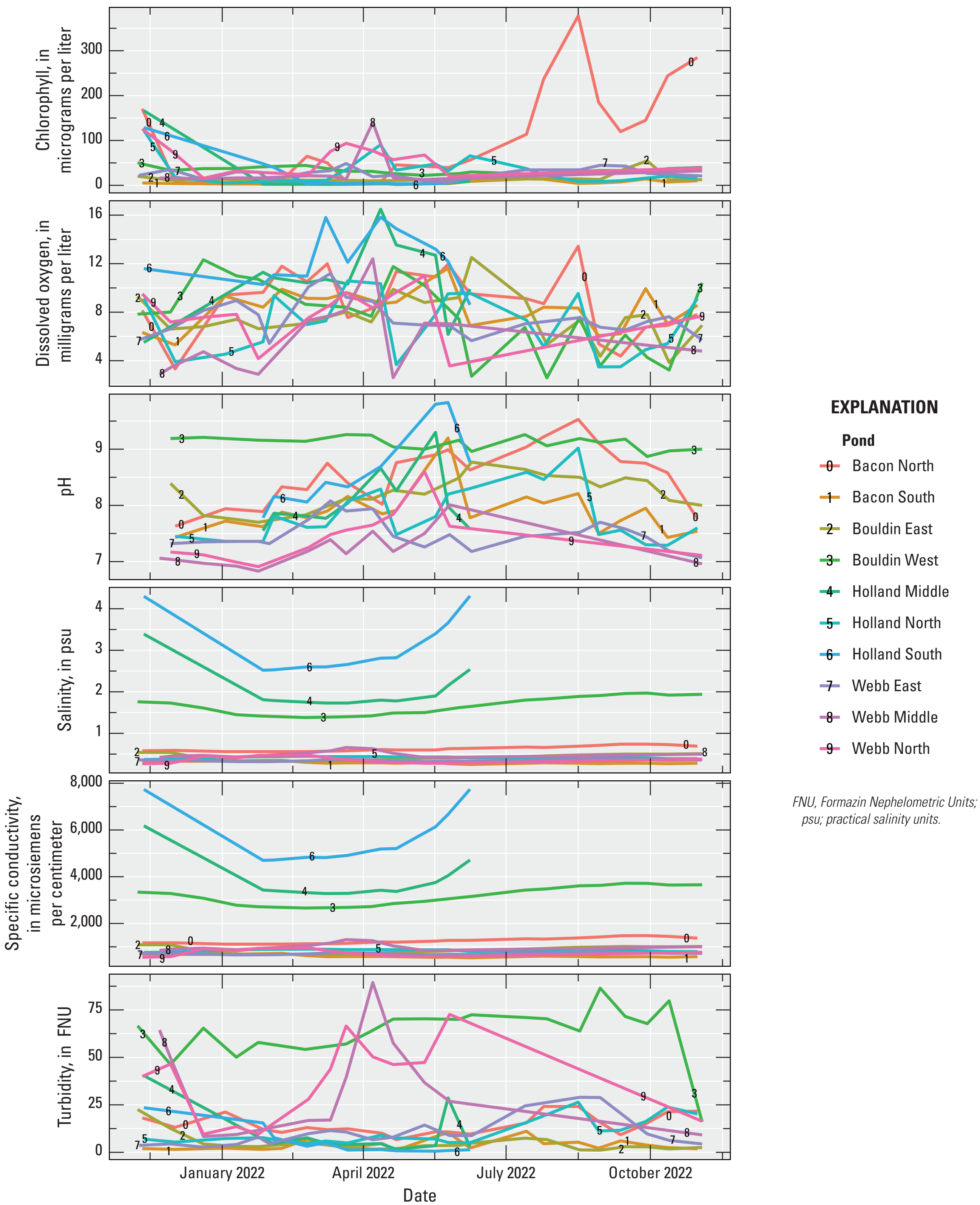

Figure 4. Water quality time series for managed ponds within the Sacramento–San Joaquin Delta in California. The Holland Middle and Holland South pond study sites do not have complete time series because they dried up during the study. Data summarized from Buxton and others (2023).

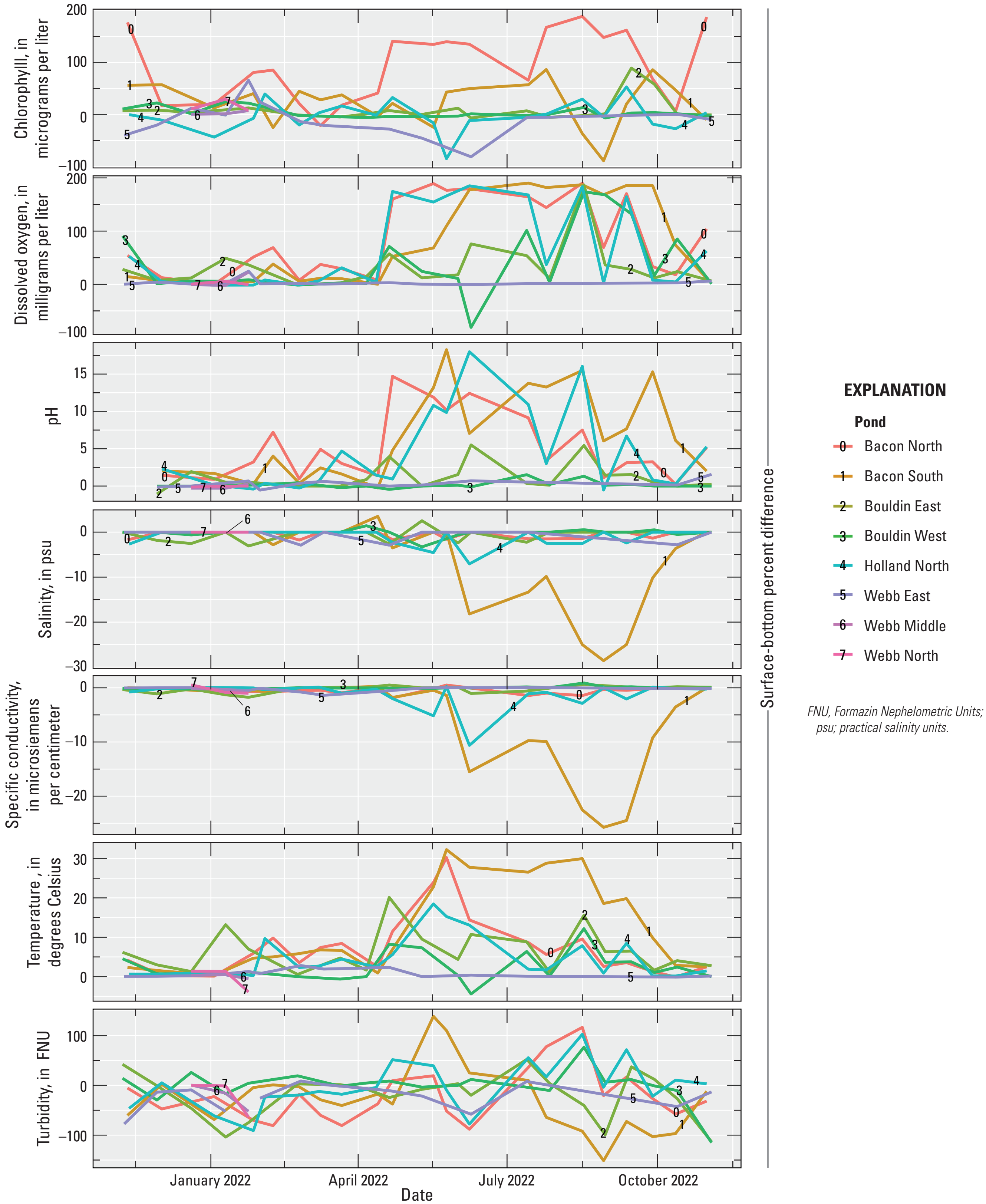

Figure 5. Water quality time series presented as the percent difference of surface relative to bottom, as an indicator of vertical stratification, for managed ponds greater than 1.5 m in depth within the Sacramento–San Joaquin Delta in California. Positive values indicate a water quality parameter showed higher values at the surface, whereas negative values indicate a water quality parameter showed higher values at the bottom. Data summarized from Buxton and others (2023).

Table 3. Results of principal components analysis (PCA) completed separately on the discrete water quality measurements (Water quality PCA) and the zooplankton species composition data (Zooplankton PCA) for managed ponds within the Sacramento–San Joaquin Delta in California. Data summarized from Buxton and others (2023).

[PC1, principal component 1; PC2, principal component 2; PC3, principal component 3]

Principal component	PC1	PC2	PC3
Water quality PCA			
Eigenvalue	1.4	1.2	1.0
Variance	33	23	17
Temperature	0.29	−0.49	0.48
Dissolved oxygen	0.34	0.58	0.30
Turbidity	0.30	−0.42	−0.67
Specific conductance	0.47	0.34	−0.40
Chlorophyll	0.29	−0.36	0.22
pH	0.63	0.02	0.12
Zooplankton PCA			
Eigenvalue	1.6	1.5	1.3
Variance	14	13	9
Arctodiaptomus dorsalis	0.31	−0.21	0.29
Diacyclops thomasi	0.37	0.23	−0.29
Acanthocyclops robustus	−0.08	0.33	0.04
Leptodiaptomus siciloides	−0.06	0.45	−0.18
Daphnia pulex	0.11	−0.21	0.06
Mesocyclops edax	0.23	−0.29	−0.08
Ostracoda	−0.37	−0.22	−0.38
Daphnia rosea	0.32	−0.20	−0.13
Daphnia magna	−0.28	−0.23	−0.43
Diaptomidae	−0.14	−0.27	−0.13
Skistodiaptomus pallidus	0.25	−0.20	−0.26
Bosmina longirostris	0.40	0.02	−0.36
Eurycercus sp.	0.12	−0.08	−0.21
Simocephalus sp.	−0.21	−0.13	−0.06
Daphnia galeata mendotae	0.17	−0.20	0.13
Acanthocyclops brevispinosus	0.06	0.27	−0.17
Diaphanosoma brachyurum	0.07	−0.06	0.33
Chydorus sphaericus	0.21	0.24	−0.16

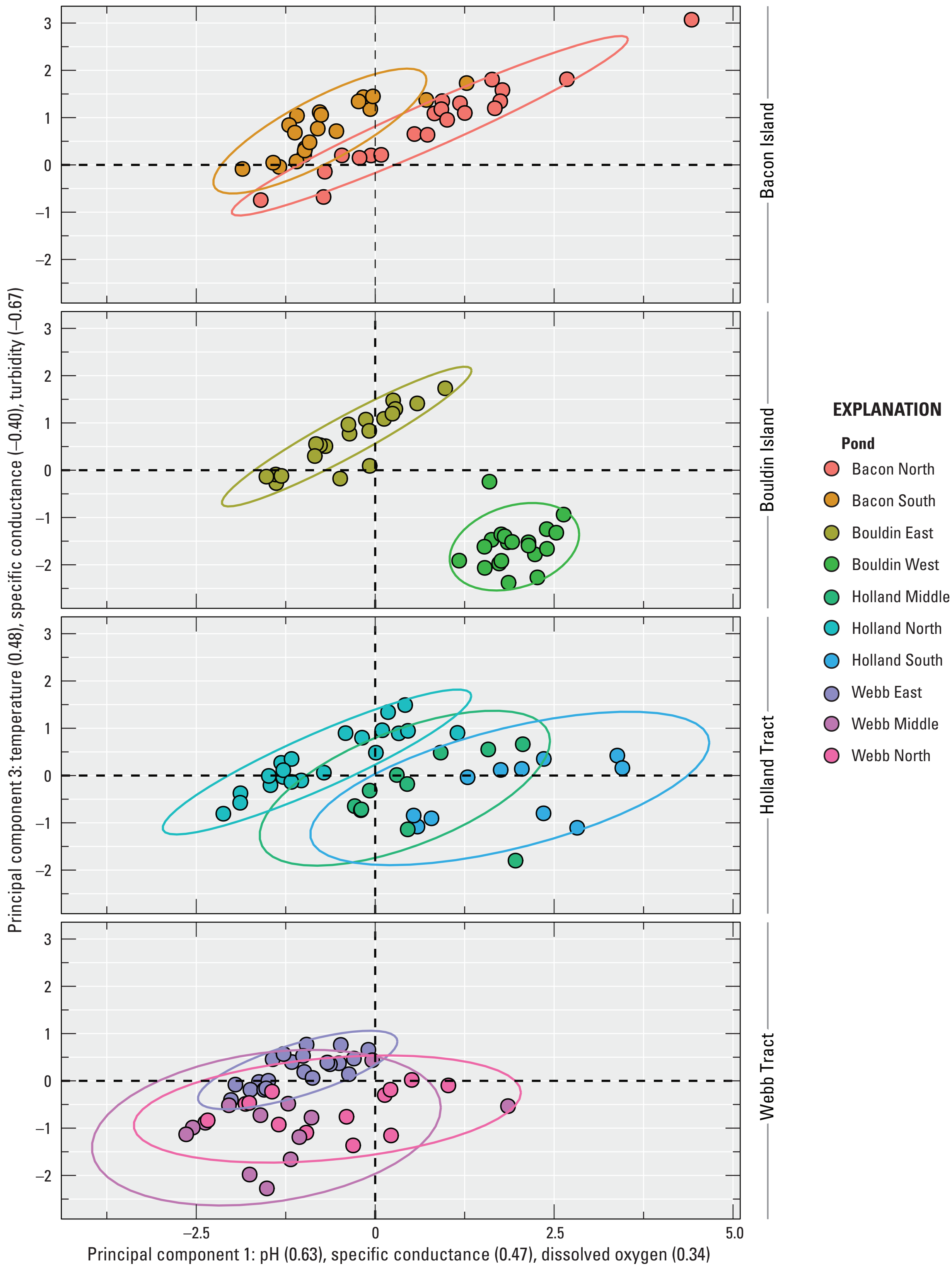

Figure 6. Scores from the first and third axes of a principal components analysis performed on discrete water quality data from managed ponds within the Sacramento–San Joaquin Delta in California. Scores are faceted by island to minimize superimposition and improve clarity. Ellipses are 95-percent confidence levels. Water quality parameters with loadings greater than 0.30 are shown for each axis. Data summarized from Buxton and others (2023).

Table 4. Water quality parameters measured in managed ponds within the Sacramento–San Joaquin Delta in California. Data summarized from Buxton and others (2023).

[Values are the average (± one standard deviation) of four measurements: (1) winter, surface, (2) winter, bottom, (3) spring, surface, (4) spring, bottom. **Abbreviations**: mg/L, milligrams per liter; µg/L, micrograms per liter; —, below detection limits; ±, plus or minus]

Parameter	Bacon Island		Bouldin Island		Holland Tract			Webb Tract		
	Bacon North	Bacon South	Bouldin East	Bouldin West	Holland Middle	Holland North	Holland South	Webb East	Webb Middle	Webb North
Nutrients										
Ammonia (mg/L)	0.354±0.14	—	0.105±0.06	0.038±0.01	0.071±0.02	1.956±1.5	0.048±0.01	—	0.079±0.01	0.068±0.01
Nitrate (mg/L)	—	—	—	—	—	—	—	—	0.029±0.03	—
Nitrite (mg/L)	—	—	—	—	0.003±0	—	—	—	0.003±0	0.003±0
Nitrogen, total (mg/L)	2.0±0.12	1.0±1.04	2.4±0.26	2.3±1.49	1.2	3.1±1.63	1.7	1.0±0.47	1.4	1.8
Organic nitrogen (mg/L)	—	—	—	—	—	—	—	—	1.3	—
Orthophosphate (mg/L)	—	—	0.542±0.62	0.345±0.26	—	0.975±0.75	—	0.082±0.01	0.242±0.17	0.103±0.07
Metals										
Barium (µg/L)	63±7.74	46±10.81	83±8.07	93±4.62	65	48±4.14	59	64±19	54	73
Calcium (mg/L)	58±4.83	31±5.64	55±9.21	28±1.7	203	35±1.18	264	30±3.35	44	51
Magnesium (mg/L)	38±4.78	21±2.68	32±5.03	120±15.6	101	22±0.99	147	19±0.45	27	30
Manganese (µg/L)	391±364.54	499±941	446±150	6±3.42	747	370±70.25	714	143±82.58	672	805
Selenium (µg/L)	0.17±0.01	—	0.15±0.02	0.18±0.04	—	—	—	—	0.09	0.14
Strontium (µg/L)	503±16.92	244±39.26	493±86	512±65.92	1,540	290±1.63	2,080	240±25.58	364	413
Zinc (µg/L)	—	—	—	—	4	—	—	—	2	—
Sediment										
Suspended (mg/L)	56±77.6	11+6.53	11±3.43	31±26.73	64±43.3	—	27±0.71	34±17.5	35±21.21	51±18.68
Organics										
Organic carbon (mg/L)	22.2±4.43	5.1±1.58	34.8±6.32	58.0±11.79	19.2	9.6±0.49	28.5	10.1±0.59	21.8	27.2
Chlorophyll *a* (µg/L)	138.2±199.7	15.8±16.2	10.9±13.28	40.6±20.29	0.4	54.7±56.39	0.7	48.7±34.3	27.1	29.0
Pheophytin *a* (µg/L)	7.9±2.8	8.8±11.47	2.3±1.05	7.8±3.01	0.9	14.6±9.5	1.1	18.0±3.68	22.3	20.0
Inorganics										
Silica (mg/L)	10±5.8	11±3.5	11±0.8	29±2.06	5	25±0.13	4	20±3.29	30	35
Hardness (mg/L)	304±8.6	164±24.56	270±43.6	564±68.5	925	177±7.23	1,270	153±10.12	221	249

Pesticides

A total of 35 pesticides were detected in water samples across all 3 surveys: 10 fungicides, 16 herbicides, and 9 insecticides (fig. 7; U.S. Geological Survey, 2024). The most frequently detected pesticides in water samples were methoxyfenozide (insecticide; 94 percent frequency), glyphosate (herbicide; 86 percent), and hexazinone (herbicide; 59 percent; U.S. Geological Survey, 2024). A total of 5 pesticides were detected in suspended sediment samples (deltamethrin [insecticide], dithiopyr [herbicide], p,p'-DDD [insecticide], p,p'-DDE [insecticide], and pendimethalin [herbicide]); though, in general, pesticide detections in suspended sediment samples were infrequent (fig. 7; U.S. Geological Survey, 2024). A total of 23 pesticides were detected in bed sediment samples (assessed only during Survey 1): 2 fungicides, 7 herbicides, 13 insecticides, and the synergist piperonyl butoxide (fig. 7; U.S. Geological Survey, 2024). The most frequently detected pesticides in bed sediment samples were p,p'-DDE (insecticide; 95 percent frequency), bifenthrin (insecticide; 75 percent), and pendimethalin (herbicide; 75 percent).

In general, more pesticides, typically in higher concentrations, were detected in the adjoining inlet/outlet of the managed ponds than in the managed ponds themselves (figs. 7, 8). In water samples, 35 pesticides were detected in adjoining inlet/outlet canals versus 25 in managed ponds. In bed sediment samples, 22 pesticides were detected in adjoining inlet/outlet canals versus 19 in managed ponds. Total combined pesticide concentrations in water samples ranged from 3.7 ng/L at Holland Middle to 6,862 ng/L at the west inlet of Webb Tract East (fig. 8). Maximum concentrations of individual pesticides detected in water samples across sites were typically less than 100 ng/L, with the exception of the herbicides diuron (530 ng/L) at Bouldin East inlet, glyphosate (6,823 ng/L) at the west inlet of Webb East, hexazinone (292 ng/L) at Bacon South inlet, pendimethalin (288.1 ng/L) at Bacon South inlet, the diuron degradate DCPMU (134.3 ng/L) at Bouldin East inlet, and the insecticide methoxyfenozide (170.6 ng/L) at Bouldin East inlet (fig. 8; U.S. Geological Survey, 2024). For nearly all samples, herbicides made up the bulk of the total amount of pesticides, which was overwhelmingly composed of glyphosate (fig. 8). Total pesticide concentrations in water were generally higher for samples collected during Survey 2, although that result was heavily affected by samples from Bouldin Island (fig. 8). Total pesticide concentrations in bed sediment samples ranged from 0.5 to 62.0 nanograms per gram (ng/g) and, in most samples, insecticides made up most of the detections (U.S. Geological Survey, 2024). Pesticide concentrations in suspended sediment samples were generally below the method reporting limit but above the method detection limit for the pesticides detected (U.S. Geological Survey, 2024). Pesticide concentrations in bed sediment samples were generally less than 2 ng/g (fig. 7; U.S. Geological Survey, 2024).

Zooplankton

A total of 46 distinct zooplankton taxa were reported across all ponds (Buxton and others, 2023). Of this total, there were 18 taxa that each made up at least 1 percent of the total biomass (table 5). Numerous rare taxa comprised less than 1 percent of the total biomass. Copepod nauplii or copepodites, which were likely not effectively sampled, were excluded from the analysis. Overall total biomass was dominated by calanoid copepods (order Calanoida; 37.4 percent), cladocerans (order Diplostraca; 28.4 percent), cyclopoid copepods (order Cyclopoida; 29.3 percent), and ostracods (class Ostracoda; 4.9 percent; table 5; fig. 9). *Arctodiaptomus dorsalis* was the dominant calanoid copepod, *Daphnia pulex* was the dominant cladoceran, and *Diacyclops thomasi* was the dominant cyclopoid copepod. Ostracods were not identified to a lower taxonomic level of resolution and, therefore, are represented by the class Ostracoda. Overall average total zooplankton biomass was 0.6 µg/L (min=0, max=63.6) and peaked during spring at over 4 µg/L (fig. 9). Results of the zooplankton PCA (table 3) indicated high overlap in species composition among all ponds (fig. 10).

Fish

Fish were reported in all managed ponds except Holland South (table 6). Among the managed ponds with fish present, the number of species reported ranged from 1 (Bouldin West, Holland Middle, and Webb Middle) to 8 (Holland North; table 6). Overall, a total of 12 individual species were reported with the most common being bluegill (*Lepomis macrochirus*) and western mosquitofish (*Gambusia affinis*), each having occurred in 5 individual managed ponds. Largemouth bass (*Micropterus salmoides*), a piscivorous species that could potentially prey on delta smelt, was present in four managed ponds, Bacon South, Bouldin East, Holland North, and Webb East. Black crappie (*Pomoxis nigromaculatus*), another piscivore and potential delta smelt predator, was present in one managed pond, Holland North. Prickly sculpin (*Cottus asper*) was the only native fish species reported and was present in three managed ponds, Bacon North, Bouldin East, and Webb.

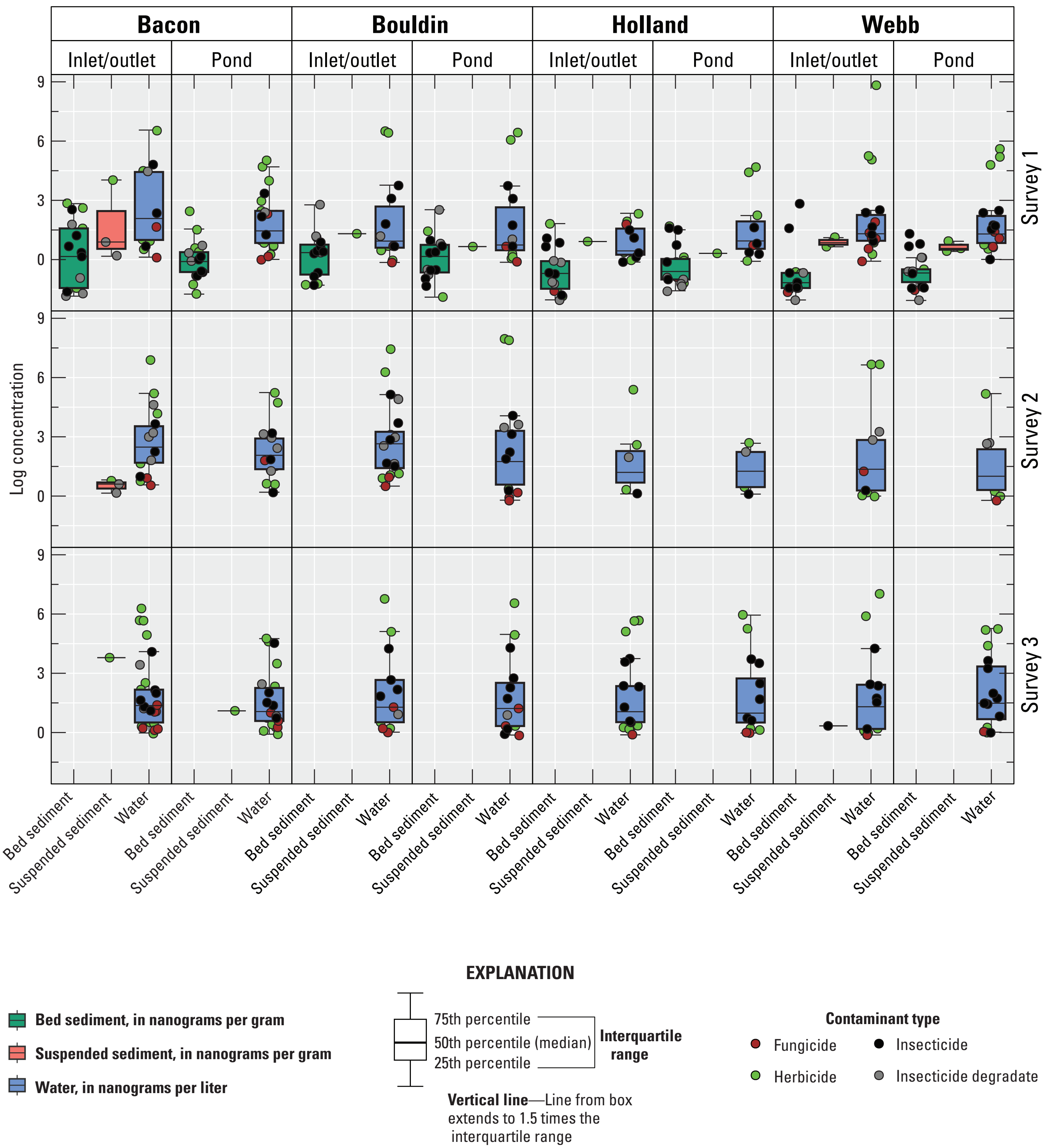

Figure 7. Pesticide concentrations in bed sediment (in nanograms per gram), suspended sediment (in nanograms per gram), and water (in nanograms per liter), by contaminant type, collected from managed ponds within the Sacramento–San Joaquin Delta in California. Concentration values are plotted on a log scale to facilitate visualization. Data summarized from U.S. Geological Survey, 2024.

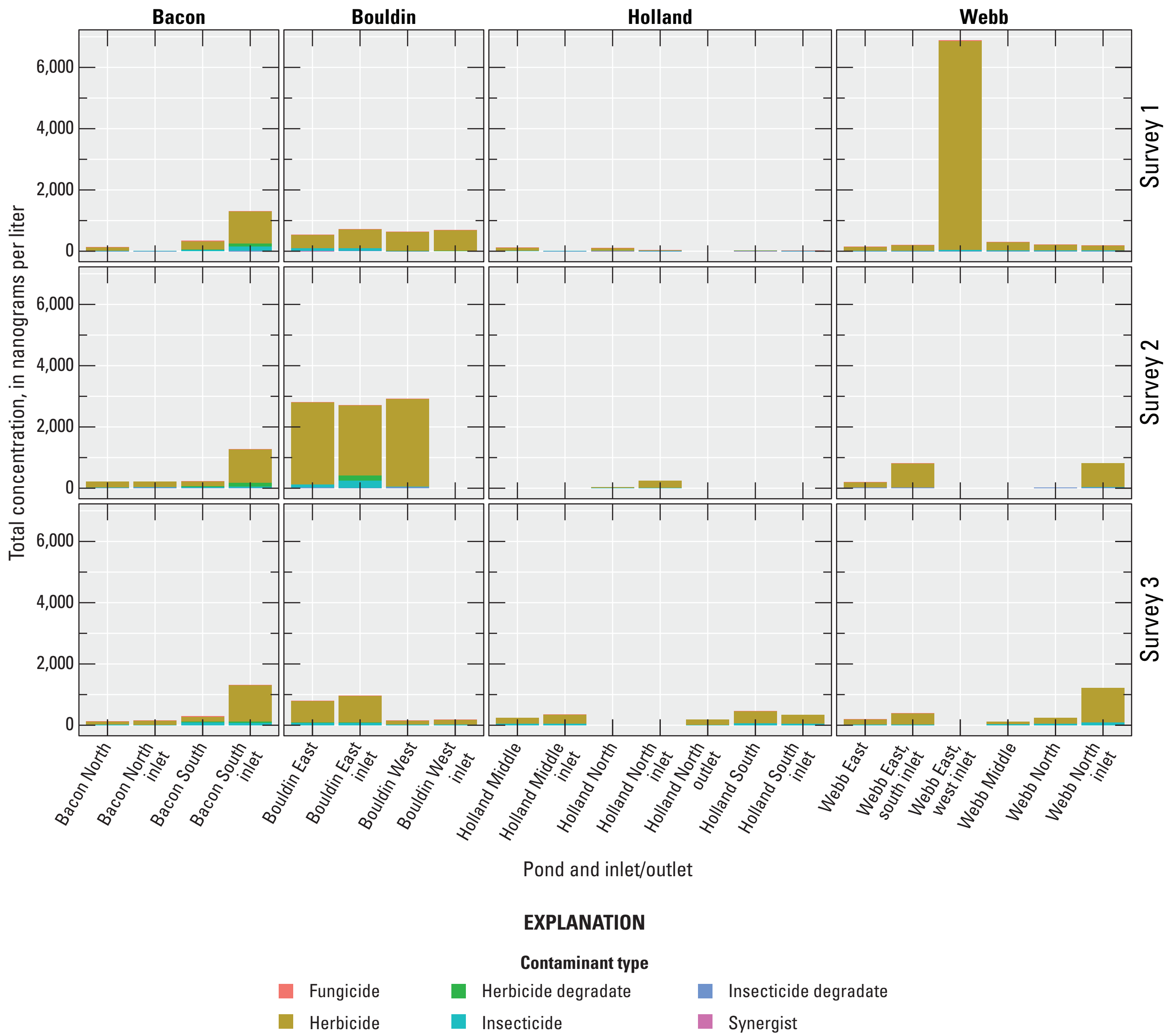

Figure 8. Total pesticide concentrations in water by contaminant type, collected from managed ponds within the Sacramento–San Joaquin Delta in California. Data summarized from U.S. Geological Survey, 2024.

Table 5. Zooplankton taxa which consisted of greater than or equal to 1 percent of the overall total biomass in managed ponds within the Sacramento–San Joaquin Delta in California. Data summarized from Buxton and others (2023).

[Ostracoda orders are not provided. **Abbreviations**: µg, microgram; —, no data]

Taxa	Order	Biomass (µg)	Percentage of total biomass
Arctodiaptomus dorsalis	Calanoida	533	20.7
Diacyclops thomasi	Cyclopoida	291	11.3
Acanthocyclops robustus	Cyclopoida	264	10.3
Leptodiaptomus siciloides	Calanoida	241	9.4
Daphnia pulex	Diplostraca	179	7.0
Mesocyclops edax	Cyclopoida	167	6.5
Ostracoda	—	125	4.9
Daphnia rosea	Diplostraca	124	4.8
Daphnia magna	Diplostraca	123	4.8
Diaptomidae	Calanoida	94	3.7
Skistodiaptomus pallidus	Calanoida	94	3.7
Bosmina longirostris	Diplostraca	74	2.9
Eurycercus sp.	Diplostraca	69	2.7
Simocephalus sp.	Diplostraca	53	2.1
Daphnia galeata mendotae	Diplostraca	51	2.0
Acanthocyclops brevispinosus	Cyclopoida	32	1.2
Diaphanosoma brachyurum	Diplostraca	29	1.1
Chydorus sphaericus	Diplostraca	28	1.1

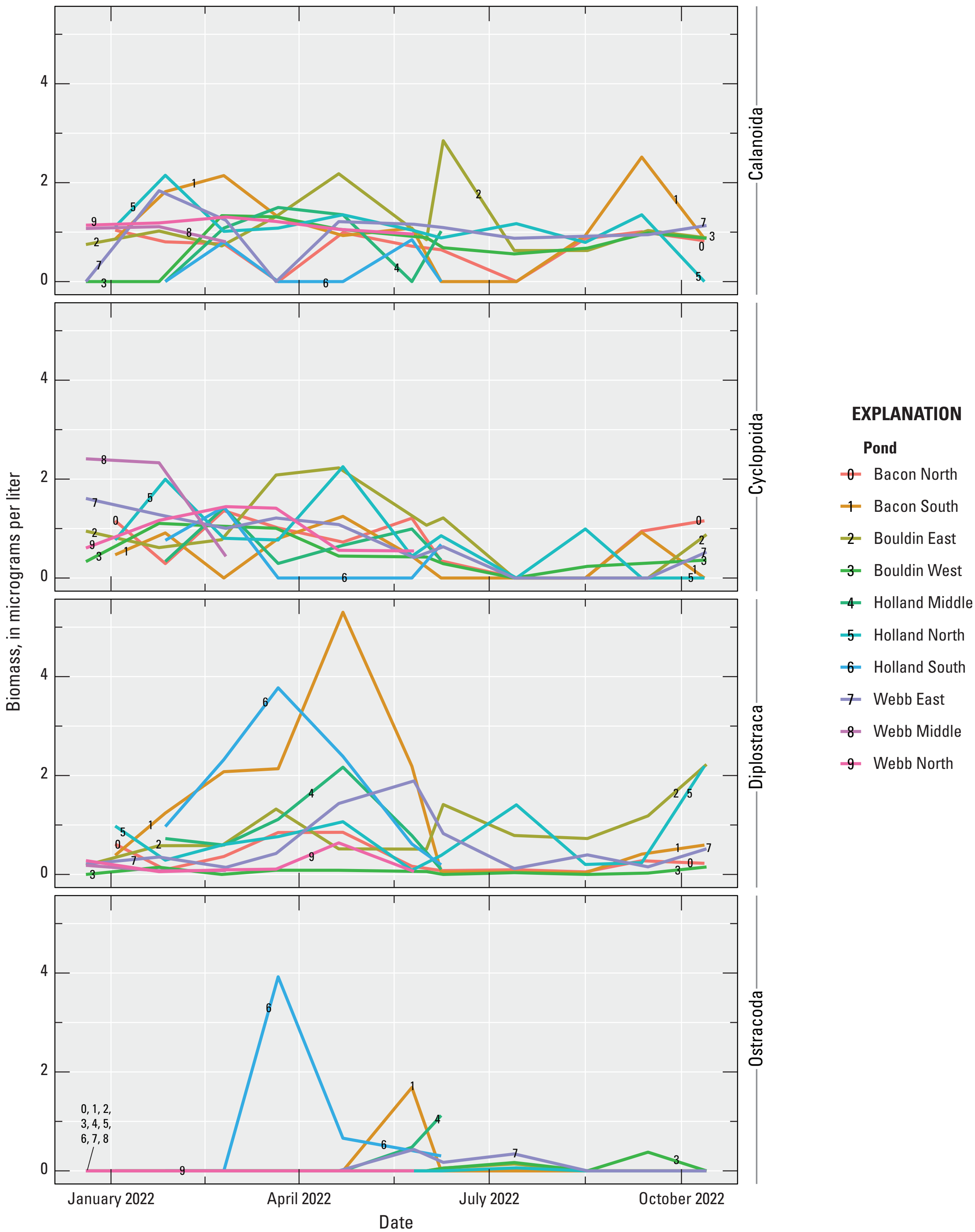

Figure 9. Zooplankton biomass time series in managed ponds within the Sacramento–San Joaquin Delta in California. Data summarized from Buxton and others (2023).

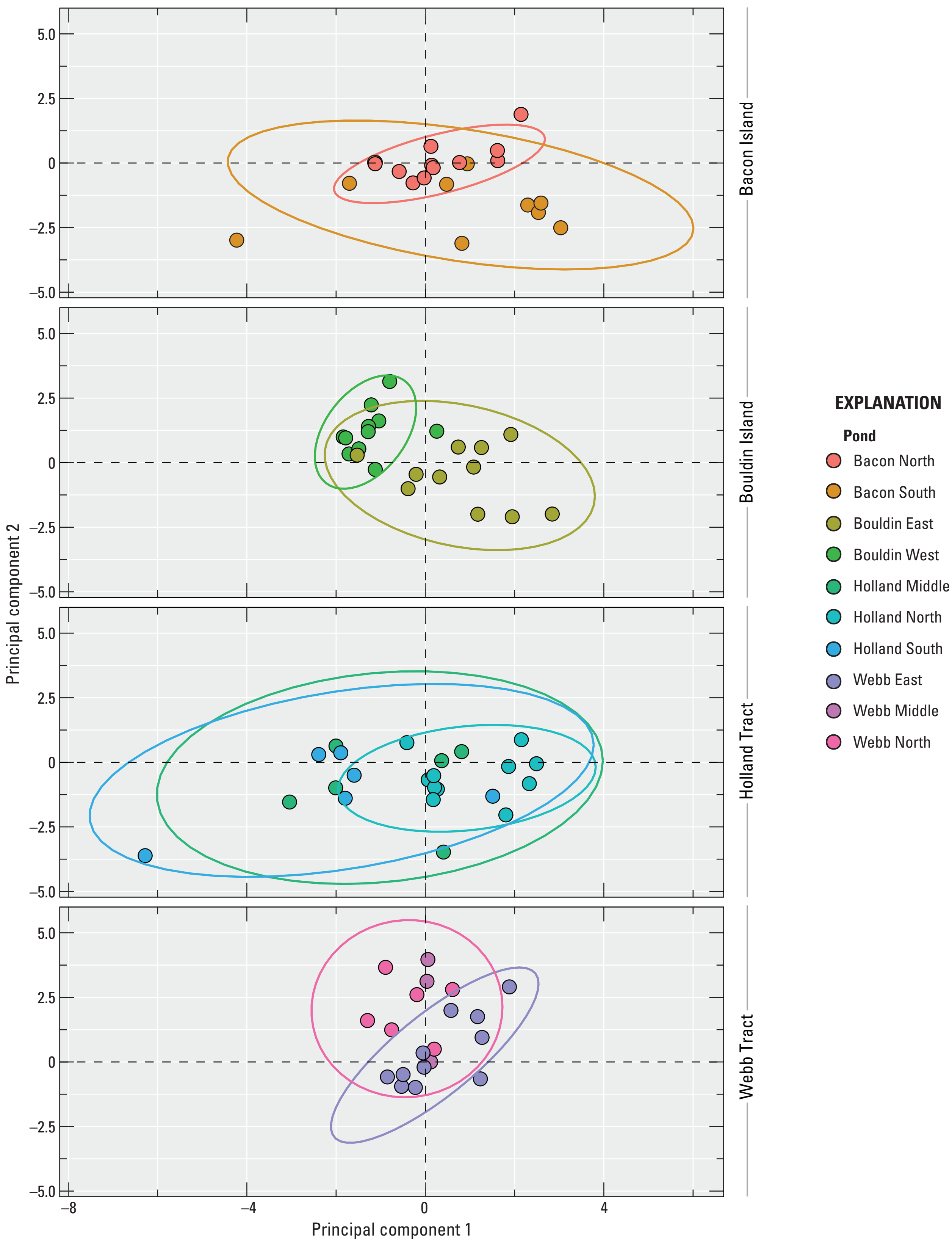

Figure 10. Scores from the first and second axes of a principal components analysis done on zooplankton species composition data from managed ponds within the Sacramento–San Joaquin Delta in California. Scores are faceted by island to minimize superimposition and improve clarity. Ellipses are 95-percent confidence levels. Data summarized from Buxton and others (2023).

Table 6. Fish species reported in managed ponds within the Sacramento–San Joaquin Delta in California. Data summarized from Buxton and others (2023).

[X, species detected; —, species not detected]

Taxa	Bacon Island		Bouldin Island		Holland Tract			Webb Tract		
	Bacon North	Bacon South	Bouldin East	Bouldin West	Holland Middle	Holland North	Holland South	Webb East	Webb Middle	Webb North
Bigscale logperch (*Percina macrolepida*)	—	—	—	—	—	X	—	—	—	—
Black Bullhead (*Ameiurus melas*)	—	—	—	—	—	—	—	X	—	X
Black Crappie (*Pomoxis nigromaculatus*)	—	—	—	—	—	X	—	—	—	—
Bluegill (*Lepomis macrochirus*)	X	X	X	—	—	X	—	—	—	X
Brown Bullhead (*Ameiurus nebulosus*)	—	—	—	—	—	X	—	—	—	—
Common Carp (*Cyprinus carpio*)	—	—	—	—	—	X	—	X	—	X
Golden Shiner (*Notemigonus crysoleucas*)	—	—	—	—	—	X	—	—	—	X
Goldfish (*Carassius auratus*)	—	—	—	—	—	X	—	—	X	X
Largemouth Bass (*Micropterus salmoides*)	—	X	X	—	—	X	—	X	—	—
Western Mosquitofish (*Gambusia affinis*)	—	X	—	X	X	—	—	X	—	X
Prickly Sculpin (*Cottus asper*)	X	—	X	—	—	—	—	X	—	—
Red Shiner (*Cyprinella lutrensis*)	—	—	—	—	—	—	—	X	—	X
Total species encountered	2	3	3	1	1	8	0	6	1	7

Discussion

Managed ponds in the Sacramento–San Joaquin Delta (Delta) of California surveyed in our study indicated a range of physical configurations, water quality characteristics, pesticide concentrations, and biological characteristics. Overall, water quality conditions in the surveyed managed ponds seemed to be driven primarily by a combination of physical habitat conditions and connectivity with adjacent managed ponds. Vertical stratification represented the dominant within-pond mode of variability in water quality conditions and was controlled by depth. Highly connected managed ponds, such as those on Webb Tract, indicated similar water quality conditions, whereas unconnected managed ponds, such as those on Bouldin Island, indicated unique water quality conditions (fig. 6). Terminal ponds with no outlets, Holland Middle and Holland South, indicated elevated salinity and high values across all non-organic water quality parameters, indicating evaporation is likely also an important driver of some aspects of water quality.

Temperature seemed to be the key physiochemical feature that may limit opportunities to support pond rearing of delta smelt. Elevated summer and fall temperatures may compromise delta smelt health and survival and thus may limit potential experimental activities to winter and spring (Komoroske and others, 2015; Jeffries and others, 2016). From May to October, the delta smelt temperature stress threshold value of 21 °C was consistently exceeded and the temperature mortality threshold value of 28 °C was intermittently exceeded (fig. 3). However, temperatures were consistently 2–5 degrees lower at the bottom of the deepest managed ponds (fig. 5), indicating that newly constructed ponds could potentially provide a deep water temperature refugia for delta smelt if properly engineered.

Pesticides were mostly detected at low concentrations but might pose a threat to delta smelt or other fishes under some specific circumstances. Measured contaminant concentrations were generally below EPA Aquatic Life Benchmarks, with just a few exceptions. Two water samples collected during Survey 2 contained pesticides at concentrations that exceeded EPA Aquatic Life Benchmarks. The Bouldin Island East Pond Inlet sample contained the herbicide diuron at a concentration of 530 ng/L, which exceeds the EPA benchmark for toxicity to vascular plants (130 ng/L). The Bouldin Island East Pond sample contained the insecticide clothianidin at a concentration of 58.6 ng/L, which exceeds the EPA benchmark for chronic toxicity to invertebrates (50 ng/L). Three water samples collected during Survey 3 contained pesticides at concentrations that exceeded EPA Aquatic Life Benchmarks. The Bouldin Island East Pond Inlet, Bouldin Island East Pond Center, and Webb Tract Middle Pond Inlet samples contained clothianidin at concentrations of 69.9, 72.8, and 70.3 ng/L, respectively, which exceeds the EPA benchmark for chronic toxicity to invertebrates. The Webb Tract East Pond South Inlet site also had a concentration of 1.4 ng/L of the pyrethroid insecticide deltamethrin in suspended sediments, which was above the EPA acute invertebrate toxicity benchmark of 0.1 ng/L. Six pyrethroid insecticides were detected in bed sediments. Specific pesticides such as fluoridone and glyphosate may be of concern because Jin and others (2018) indicated that fluoridone and glyphosate affected liver estradiol hormone and reduced oxidative enzyme, and fluoridone could affect brain activity in delta smelt. Additionally, Jeffries and others (2015) indicated that permethrin activated genes for protein synthesis and metabolism, and resulted in mortality in delta smelt, but the study used 100 times the concentrations of pesticides that were detected here in this study.

The surveyed managed ponds were all highly productive and seemed to possess food webs that could support delta smelt in terms of type and quantity of available prey. Chlorophyll *a* concentration, an index of potential primary productivity, indicated the managed ponds were productive, with several managed ponds indicating eutrophic (approximately 10–40 µg/L) or hypereutrophic (greater than 40 µg/L) conditions. Zooplankton populations were broadly similar among managed ponds and included calanoid and cyclopoid copepods that would be suitable prey for delta smelt. Additional research will be needed to estimate the carrying capacity of managed pond food webs to support delta smelt and other native fishes and to determine the potential uptake and bioaccumulation of pesticides (of high concentrations) in zooplankton or key food items of delta smelt.

Managed ponds may support fish species that could potentially prey on delta smelt. Potential predators such as largemouth bass and black crappie were reported in 4 and 1 of the surveyed managed ponds, respectively. The distribution of predators among ponds seemed to have been driven primarily by deliberate stocking to facilitate local fisheries. Overall, the surveyed pond fish communities were unnatural assemblages of species that did not mirror assemblages in adjacent sloughs (Feyrer and Healey, 2003; Nobriga and others, 2005), indicating that sportfish stocking and water management practices controlled fish occupancy.

Overall, the managed ponds surveyed in our study seem to be suitable to support delta smelt and other native fishes, at least during some parts of the year. Indeed, delta smelt experimentally reared during winter 2022–23 in Bouldin East and Bouldin West indicated high survival and good health (Shawn Acuña, Metropolitan Water District, unpub. data, 2025). The biggest challenge to successfully rearing delta smelt in managed ponds in the Delta may be the high summer water temperature. However, this could potentially be mitigated by managing pond depth and water circulation to provide thermal refugia and suitable water quality conditions. Properly engineered and managed ponds seem to hold great promise for research and conservation efforts involving delta smelt and other native fishes.

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Appendix 1. Water Quality Survey Sample Sites

Table 1.1. Water quality survey sample sites in managed ponds within the Sacramento–San Joaquin Delta in California (U.S. Geological Survey, 2024).

[USGS, U.S. Geological Survey]

Managed pond	USGS site identifier	Latitude	Longitude
Bacon North	375955121333601	37.9986	−121.5601
Bacon South	375734121330601	37.9594	−121.5517
Bouldin East	380545121310901	38.0957	−121.5191
Bouldin West	380557121323001	38.0992	−121.5416
Holland Middle	380053121353101	38.0147	−121.5920
Holland North	380137112135201	38.0269	−121.5866
Holland South	380052121354001	38.0145	−121.5945
Webb East	380502121350301	38.0840	−121.5843
Webb Middle	380543121362801	38.0952	−121.6077
Webb North	380551121364201	38.0976	−121.6118

Reference Cited

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Appendix 2. Pesticide Survey Sample Sites

Table 2.1. Pesticide survey sample sites in managed ponds within the Sacramento–San Joaquin Delta in California (U.S. Geological Survey, 2024).

[USGS, U.S. Geological Survey]

Managed pond	Location	USGS site identifier	Latitude	Longitude
Bacon North	Inlet	375957121332901	37.99934	−121.55818
Bacon North	Center	375955121333601	37.99861	−121.56013
Bacon North	Inlet	375957121332901	37.96121	−121.55250
Bacon North	Center	375955121333601	37.95940	−121.55167
Bouldin East	Inlet	380546121311501	38.09627	−121.52101
Bouldin East	Center	380545121310901	38.09573	−121.51909
Bouldin West	Inlet	380600121322501	38.10010	−121.54028
Bouldin West	Center	380557121323001	38.09924	−121.54156
Holland Middle	Inlet	380052121353601	38.01459	−121.59347
Holland Middle	Center	380053121353101	38.01472	−121.59196
Holland North	Inlet	380141121351401	38.02809	−121.58744
Holland North	Center	380137112135201	38.02692	−121.58658
Holland South	Inlet	380051121353701	38.01432	−121.59367
Holland South	Center	380052121354001	37.01447	−121.59449
Holland North	Outlet	380124121351201	38.02330	−121.58667
Webb East	Center	380502121350301	38.08395	−121.58427
Webb East	Inlet	380451121350701	38.08089	−121.58534
Webb East	Inlet (West)	380506121352801	38.08511	−121.59123
Webb North	Inlet (Middle)	380527121363601	38.09105	−121.61012
Webb Middle	Center	380543121362801	38.09517	−121.60768
Webb North	Center	380551121364201	38.09756	−121.61179

Reference Cited

U.S. Geological Survey, 2024, USGS water data for the Nation: U.S. Geological Survey National Water Information System database, accessed January 24, 2025, at https://doi.org/10.5066/F7P55KJN.

Appendix 3. Method Detection and Reporting Limits for Pesticides Dissolved in Water and Sediments Measured by the U.S. Geological Survey Organic Chemistry Research Laboratory

Table 3.1. Method detection and reporting limits for pesticides dissolved in water and sediments measured by the U.S. Geological Survey Organic Chemistry Research Laboratory.

[CAS, chemical abstracts service; GC/MS/MS, gas chromatography with tandem mass spectrometry; LC/MS/MS, liquid chromatography with tandem mass spectrometry; MDL, method detection limit; NA, not analyzed; ng/g, nanograms per gram; ng/L, nanograms per liter; NWIS, National Water Information System; RL, reporting limit]

Compound	CAS number	Chemical class	Pesticide type	NWIS water and suspended sediment parameter code	Water RL (ng/L)	Water MDL (ng/L)	Suspended sediment RL (ng/L)	Suspended sediment MDL (ng/L)	NWIS sediment parameter code	Bed sediment RL (ng/g)	Bed sediment MDL (ng/g)	Analytical instrument
Acetamiprid	135410-20-7	Neonicotinoid	Insecticide	68302	2.1	1	4.4	2.2	54365	0.209453	0.104726	LC/MS/MS
Acetochlor	34256-82-1	Chloroacetanilide	Herbicide	68520	3.1	1.5	3.4	1.7	54366	0.412551	0.206276	LC/MS/MS
Acibenzolar-S-methyl	135158-54-2	Benzothiadiazole	Fungicide	51849	10.7	5.3	11.1	5.6	51870	0.794744	0.397372	GC/MS/MS
Allethrin	584-79-2	Pyrethroid	Insecticide	66586	3.8	1.9	6.2	3.1	66588	0.807834	0.403917	GC/MS/MS
Atrazine	1912-24-9	Triazine	Herbicide	65065	1.7	0.9	2.7	1.4	39631	0.210395	0.105198	LC/MS/MS
Atrazine, Desethyl	6190-65-4	Triazine	Herbicide degradate	68552	3.2	1.6	4.5	2.3	4001	0.401855	0.200928	LC/MS/MS
Atrazine, Desisopropyl	1007-28-9	Triazine	Herbicide degradate	68550	3.7	1.8	5.6	2.8	4003	0.412079	0.206039	LC/MS/MS
Azoxystrobin	131860-33-8	Strobin	Fungicide	66589	1.6	0.8	4.3	2.2	66591	0.200928	0.100464	LC/MS/MS
Benefin (Benfluralin)	1861-40-1	2,6-Dinitroaniline	Herbicide	51643	3.6	1.8	6.8	3.4	68878	0.196086	0.098043	GC/MS/MS
Bentazon	25057-89-0	Benzothiadiazine	Herbicide	68538	2.5	1.3	NA	NA	54421	0.402597	0.201299	LC/MS/MS
Benzobicyclon	156963-66-5	Carbobicyclic	Herbicide	54350	2.3	1.2	3.5	1.8	54424	0.470438	0.235219	LC/MS/MS
Benzovindiflupyr	1072957-71-1	Amide	Fungicide	52652	2.3	1.2	3.6	1.8	54367	0.227841	0.11392	LC/MS/MS
Bifenthrin	82657-04-3	Pyrethroid	Insecticide	65067	1.1	0.6	1.5	0.8	64151	0.212519	0.10626	GC/MS/MS
Boscalid	188425-85-6	Anilide	Fungicide	67550	2	1	3.5	1.7	67552	0.432196	0.216098	LC/MS/MS
Boscalid Metabolite - M510F01 Acetyl	661463-87-2	Anilide	Fungicide degradate	54349	1.6	0.8	3.3	1.7	54423	0.26395	0.131975	LC/MS/MS
Broflanilide	1207727-04-5	Benzamide	Insecticide	54363	3.9	1.9	4.2	2.1	54445	0.262166	0.131083	LC/MS/MS
Bromoxynil	1689-84-5	Nitrile	Herbicide	NA	NA	NA	NA	NA	54434	0.54	0.27	LC/MS/MS
Bromoconazole	116255-48-2	Azole	Fungicide	68315	1.9	1	3.8	1.9	68317	0.419085	0.209542	LC/MS/MS
Butralin	33629-47-9	2,6-Dinitroaniline	Herbicide	68545	2.5	1.2	3.6	1.8	68880	0.404781	0.20239	LC/MS/MS
Carbaryl	63-25-2	N-Methyl Carbamate	Insecticide	65069	1.7	0.8	3.5	1.7	64153	0.207572	0.103786	LC/MS/MS
Carbendazim	10605-21-7	Benzimidazole	Fungicide	68548	2.5	1.2	4.9	2.5	NA	NA	NA	LC/MS/MS
Carbofuran	1563-66-2	N-Methyl Carbamate	Insecticide	65070	1.3	0.6	3.1	1.5	64154	0.204415	0.102208	LC/MS/MS
Chlorantraniliprole	500008-45-7	Anthranilic diamide	Insecticide	51856	1.5	0.7	3.7	1.8	54370	0.213412	0.106706	LC/MS/MS
Chlorfenapyr	122453-73-0	Pyrrole	Insecticide	53567	3.6	1.8	5	2.5	54447	0.858716	0.429358	GC/MS/MS

Table 3.1. Method detection and reporting limits for pesticides dissolved in water and sediments measured by the U.S. Geological Survey Organic Chemistry Research Laboratory.—Continued

[CAS, chemical abstracts service; GC/MS/MS, gas chromatography with tandem mass spectrometry; LC/MS/MS, liquid chromatography with tandem mass spectrometry; MDL, method detection limit; NA, not analyzed; ng/g, nanograms per gram; ng/L, nanograms per liter; NWIS, National Water Information System; RL, reporting limit]

Compound	CAS number	Chemical class	Pesticide type	NWIS water and suspended sediment parameter code	Water RL (ng/L)	Water MDL (ng/L)	Suspended sediment RL (ng/L)	Suspended sediment MDL (ng/L)	NWIS sediment parameter code	Bed sediment RL (ng/g)	Bed sediment MDL (ng/g)	Analytical instrument
Chlorothalonil	1897-45-6	Substituted Benzene	Fungicide	65071	1.9	0.7	18	9	NA	NA	NA	GC/MS/MS
Chlorpyrifos	2921-88-2	Organophosphorus	Insecticide	65072	2.4	1.2	3.9	1.9	81404	0.417854	0.208927	LC/MS/MS
Chlorpyrifos oxon	5598-15-2	Organophosphorus	Insecticide	68216	2	1	3.9	2	68218	0.239709	0.119854	LC/MS/MS
Clomazone	81777-89-1	Oxazolidinone	Herbicide	67562	2.4	1.2	3.6	1.8	67564	0.19082	0.09541	LC/MS/MS
Clothianidin	210880-92-5	Neonicotinoid	Insecticide	68221	2	1	5.7	2.8	68223	0.198857	0.099428	LC/MS/MS
Clothianidin des methyl	135018-15-4	Neonicotinoid	Insecticide degradate	52660	3.7	1.8	5.6	2.8	54408	0.482081	0.241041	LC/MS/MS
Coumaphos	56-72-4	Organophosphorus	Insecticide	51836	2.3	1.1	3.7	1.8	68882	0.505642	0.252821	LC/MS/MS
Cyantraniliprole	736994-63-1	Anthranilic diamide	Insecticide	51862	2.2	1.1	3.9	2	54372	0.466417	0.233208	LC/MS/MS
Cyazofamid	120116-88-3	Azole	Fungicide	51853	1.7	0.8	3.6	1.8	54373	0.192207	0.096103	LC/MS/MS
Cyclaniliprole	1031756-98-5	Anthranilic diamide	Insecticide	54355	2.7	1.4	2.9	1.4	54435	0.548125	0.274063	LC/MS/MS
Cycloate	1134-23-2	Thiocarbamate	Herbicide	65073	1.8	0.9	3.4	1.7	64155	0.446684	0.223342	LC/MS/MS
Cyfluthrin	68359-37-5	Pyrethroid	Insecticide	65074	1.7	0.8	2.1	1	65109	0.188959	0.094479	GC/MS/MS
Cyhalofop-butyl	122008-85-9	Aryloxyphenoxy propionic acid	Herbicide	68360	3	1.5	4.4	2.2	68884	0.214013	0.107006	GC/MS/MS
Cyhalothrin (all isomers)	68085-85-8	Pyrethroid	Insecticide	68354	1.2	0.6	1.9	1	68356	0.215611	0.107806	GC/MS/MS
Cymoxanil	57966-95-7	Urea	Fungicide	51861	4.6	2.3	4.3	2.2	54374	0.436771	0.218386	LC/MS/MS
Cypermethrin	52315-07-8	Pyrethroid	Insecticide	65075	1.8	0.9	2.2	1.1	64156	0.186652	0.093326	GC/MS/MS
Cyproconazole	94361-06-5	Azole	Fungicide	66593	2.8	1.4	3.8	1.9	66595	0.208087	0.104043	LC/MS/MS
Cyprodinil	121552-61-2	Pyrimidine	Fungicide	67574	4.3	2.1	3.2	1.6	NA	NA	NA	LC/MS/MS
DCPA	1861-32-1	Alkyl Phthalate	Herbicide	65076	2.3	1.2	2.5	1.2	62905	0.203906	0.101953	GC/MS/MS
DCPMU	3567-62-2	Urea	Herbicide degradate	68231	1.5	0.7	2.6	1.3	NA	NA	NA	LC/MS/MS
DCPU	155998	Urea	Herbicide degradate	68226	2.1	1.1	3.5	1.7	NA	NA	NA	LC/MS/MS
Deltamethrin	52918-63-5	Pyrethroid	Insecticide	65077	1.4	0.7	2.8	1.4	65110	0.397768	0.198884	GC/MS/MS
Desthio-prothioconazole	120983-64-4	Azole	Fungicide degradate	51865	1.3	0.7	2.8	1.4	54375	0.215031	0.107516	LC/MS/MS
Desulfinylfipronil	205650-65-3	Pyrazole	Insecticide degradate	66607	1.9	1	2.1	1	68891	0.201768	0.100884	LC/MS/MS

Table 3.1. Method detection and reporting limits for pesticides dissolved in water and sediments measured by the U.S. Geological Survey Organic Chemistry Research Laboratory.—Continued

[CAS, chemical abstracts service; GC/MS/MS, gas chromatography with tandem mass spectrometry; LC/MS/MS, liquid chromatography with tandem mass spectrometry; MDL, method detection limit; NA, not analyzed; ng/g, nanograms per gram; ng/L, nanograms per liter; NWIS, National Water Information System; RL, reporting limit]

Compound	CAS number	Chemical class	Pesticide type	NWIS water and suspended sediment parameter code	Water RL (ng/L)	Water MDL (ng/L)	Suspended sediment RL (ng/L)	Suspended sediment MDL (ng/L)	NWIS sediment parameter code	Bed sediment RL (ng/g)	Bed sediment MDL (ng/g)	Analytical instrument
Desulfinylfipronil Amide	1115248-09-3	Pyrazole	Insecticide degradate	68570	2.1	1	2.4	1.2	66609	0.43457	0.217285	LC/MS/MS
Diazinon	333-41-5	Organophosphorus	Insecticide	65078	2.3	1.1	3.3	1.6	39571	0.211199	0.1056	LC/MS/MS
Diazoxon	962-58-3	Organophosphorus	Insecticide degradate	68236	1.5	0.7	4.1	2.1	68238	0.205959	0.102979	LC/MS/MS
3,4-Dichloroaniline	95-76-1	Amine	Herbicide degradate	66584	2.3	1.2	2.5	1.2	66585	0.95924	0.47962	LC/MS/MS
3,5-Dichloroaniline	626-43-7	Amine	Herbicide degradate	67536	5.6	2.8	5.9	3	67538	0.827873	0.413936	LC/MS/MS
Dichlorvos	62-73-7	Organophosphorus	Insecticide	68572	2.4	1.2	1.8	0.9	54376	0.439189	0.219594	LC/MS/MS
Difenoconazole	119446-68-3	Azole	Fungicide	67582	2.7	1.3	2.8	1.4	67584	0.396093	0.198047	LC/MS/MS
Dimethomorph	110488-70-5	Morpholine	Fungicide	68373	1.4	0.7	5.5	2.8	68375	0.198899	0.09945	LC/MS/MS
Dinotefuran	165252-70-0	Neonicotinoid	Insecticide	68379	3.6	1.8	7.3	3.6	54377	0.411789	0.205895	LC/MS/MS
Dithiopyr	97886-45-8	Pyridinecarboxylic acid	Herbicide	51837	2.3	1.1	2.5	1.3	68886	0.203937	0.101969	GC/MS/MS
Diuron	330-54-1	Urea	Herbicide	66598	1.4	0.7	3.8	1.9	66600	0.265939	0.132969	LC/MS/MS
EPTC	759-94-4	Thiocarbamate	Herbicide	65080	2.6	1.3	2.8	1.4	64158	0.880311	0.440156	LC/MS/MS
Esfenvalerate	66230-04-4	Pyrethroid	Insecticide	65081	1.5	0.7	2.4	1.2	64159	0.217059	0.108529	GC/MS/MS
Ethaboxam	162650-77-3	Aromatic Amide	Fungicide	51855	3	1.5	3.5	1.7	54378	0.373235	0.186617	LC/MS/MS
Ethalfluralin	55283-68-6	2,6-Dinitroaniline	Herbicide	65082	5.4	2.7	6.2	3.1	64160	0.211599	0.105799	GC/MS/MS
Etofenprox	80844-07-1	Pyrethroid Ether	Insecticide	67604	3.8	1.9	3.4	1.7	67606	0.193598	0.096799	GC/MS/MS
Etoxazole	153233-91-1	Diphenyl Oxazoline	Insecticide	68598	2.4	1.2	3.7	1.9	54379	0.256658	0.128329	LC/MS/MS
Famoxadone	131807-57-3	Oxazolidinedione	Fungicide	67609	13.9	6.9	18	9	NA	NA	NA	LC/MS/MS
Fenamidone	161326-34-7	Imidazole	Fungicide	51848	1.7	0.9	1.9	1	51869	0.215952	0.107976	LC/MS/MS
Fenbuconazole	114369-43-6	Azole	Fungicide	67618	1.8	0.9	2.9	1.5	67620	0.212074	0.106037	LC/MS/MS
Fenhexamid	126833-17-8	Anilide	Fungicide	67622	17.8	8.9	20.8	10.4	67624	5.114847	2.557423	LC/MS/MS
Fenpropathrin	39515-41-8	Pyrethroid	Insecticide	65083	1.6	0.8	3.3	1.7	65111	0.443863	0.221931	GC/MS/MS
Fenpyroximate	134098-61-6	Pyrazole	Insecticide	51838	2.8	1.4	4.3	2.2	NA	NA	NA	LC/MS/MS
Fipronil	120068-37-3	Pyrazole	Insecticide	66604	1.8	0.9	2.4	1.2	66606	0.214211	0.107106	LC/MS/MS
Fipronil sulfide	120067-83-6	Pyrazole	Insecticide degradate	66610	1.5	0.7	1.9	1	66612	0.196602	0.098301	LC/MS/MS

Table 3.1. Method detection and reporting limits for pesticides dissolved in water and sediments measured by the U.S. Geological Survey Organic Chemistry Research Laboratory.—Continued

[CAS, chemical abstracts service; GC/MS/MS, gas chromatography with tandem mass spectrometry; LC/MS/MS, liquid chromatography with tandem mass spectrometry; MDL, method detection limit; NA, not analyzed; ng/g, nanograms per gram; ng/L, nanograms per liter; NWIS, National Water Information System; RL, reporting limit]

Compound	CAS number	Chemical class	Pesticide type	NWIS water and suspended sediment parameter code	Water RL (ng/L)	Water MDL (ng/L)	Suspended sediment RL (ng/L)	Suspended sediment MDL (ng/L)	NWIS sediment parameter code	Bed sediment RL (ng/g)	Bed sediment MDL (ng/g)	Analytical instrument
Fipronil sulfone	120068-36-2	Pyrazole	Insecticide degradate	66613	1.7	0.9	2.4	1.2	66615	0.207212	0.103606	LC/MS/MS
Flonicamid	158062-67-0	Pyridinecarboxamide	Insecticide	51858	1.5	0.8	5	2.5	54380	0.436773	0.218387	LC/MS/MS
Florpyrauxifen-Benzyl	1390661-72-9	Aminopyridine	Herbicide	54356	3.1	1.5	3.3	1.7	54436	0.245081	0.122541	LC/MS/MS
Fluazinam	79622-59-6	2,6-Dinitroaniline	Fungicide	67636	2.4	1.2	2.8	1.4	67638	0.207404	0.103702	LC/MS/MS
Flubendiamide	272451-65-7	Organofluorine	Insecticide	NA	NA	NA	NA	NA	54381	0.52	0.26	LC/MS/MS
Fludioxonil	131341-86-1	Benzodioxole	Fungicide	67640	2.4	1.2	2.7	1.3	NA	NA	NA	LC/MS/MS
Flufenacet	142459-58-3	Anilide	Herbicide	51840	3.7	1.8	3.8	1.9	68893	0.201474	0.100737	LC/MS/MS
Fluindapyr	1383809-87-7	Pyrazole	Fungicide	54362	2.7	1.3	3.2	1.6	54444	0.204222	0.102111	LC/MS/MS
Flumetralin	62924-70-3	2,6-Dinitroaniline	Plant growth regulator	51841	3.4	1.7	3.8	1.9	68895	0.528529	0.264264	LC/MS/MS
Fluopicolide	239110-15-7	Benzamide Pyridine	Fungicide	51852	1.6	0.8	3.8	1.9	51873	0.197246	0.098623	LC/MS/MS
Fluopyram	658066-35-4	Amide	Fungicide	52646	1.5	0.8	3.6	1.8	54402	0.197873	0.098936	LC/MS/MS
Fluoxastrobin	193740-76-0	Strobin	Fungicide	67645	2.8	1.4	3.8	1.9	67647	0.199347	0.099674	LC/MS/MS
Flupyradifurone	951659-40-8	Butenolides	Insecticide	52764	1.4	0.7	3.3	1.7	54382	0.221827	0.110914	LC/MS/MS
Fluridone	59756-60-4	Phenylpyridine	Herbicide	51864	2.9	1.5	4.2	2.1	54383	0.199638	0.099819	LC/MS/MS
Flutolanil	66332-96-5	Anilide	Fungicide	51842	2.6	1.3	3.7	1.9	68897	0.194805	0.097403	LC/MS/MS
Flutriafol	76674-21-0	Azole	Fungicide	67653	2.7	1.4	3.8	1.9	67655	0.202849	0.101425	LC/MS/MS
Fluxapyroxad	907204-31-3	Anilide, Pyrazole	Fungicide	51851	1.4	0.7	3.4	1.7	51872	0.222944	0.111472	LC/MS/MS
Fomesafen	72178-02-0	Diphenylether	Herbicide	NA	NA	NA	NA	NA	54437	0.9	0.45	LC/MS/MS
Halauxifen-methyl ester	943831-98-9	Methyl Ester	Herbicide	54361	1.4	0.7	2.2	1.1	54442	0.200264	0.100132	LC/MS/MS
Hexazinone	51235-04-2	Triazinone	Herbicide	65085	1.2	0.6	3.3	1.7	64161	0.207321	0.103661	LC/MS/MS
Imazalil	35554-44-0	Azole	Fungicide	67662	3	1.5	NA	NA	NA	NA	NA	LC/MS/MS
Imidacloprid	138261-41-3	Neonicotinoid	Insecticide	68426	2	1	2.1	1	68428	0.207148	0.103574	LC/MS/MS
Imidacloprid desnitro	127202-53-3	Neonicotinoid	Insecticide degradate	51857	7.4	3.7	10.8	5.4	NA	NA	NA	LC/MS/MS
Imidacloprid Olefin	115086-54-9	Neonicotinoid	Insecticide degradate	52782	6.6	3.3	11	5.5	54413	0.522177	0.261089	LC/MS/MS
Imidacloprid Urea	120868-66-8	Neonicotinoid	Insecticide degradate	51859	2.8	1.4	4	2	54384	0.261432	0.130716	LC/MS/MS

Table 3.1. Method detection and reporting limits for pesticides dissolved in water and sediments measured by the U.S. Geological Survey Organic Chemistry Research Laboratory.—Continued

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Compound	CAS number	Chemical class	Pesticide type	NWIS water and suspended sediment parameter code	Water RL (ng/L)	Water MDL (ng/L)	Suspended sediment RL (ng/L)	Suspended sediment MDL (ng/L)	NWIS sediment parameter code	Bed sediment RL (ng/g)	Bed sediment MDL (ng/g)	Analytical instrument
5-OH Imidacloprid	380912-09-4	Neonicotinoid	Insecticide degradate	54344	4.1	2	4.4	2.2	54415	0.391162	0.195581	LC/MS/MS
Indaziflam	950782-86-2	Alkylazine	Herbicide	53960	2.5	1.3	4	2	54403	0.213803	0.106902	LC/MS/MS
Indoxacarb	173584-44-6	Oxadiazine	Insecticide	68627	3.2	1.6	3.5	1.7	68899	0.409951	0.204975	LC/MS/MS
Ipconazole	125225-28-7	Triazole	Fungicide	52762	2.4	1.2	4.1	2.1	54385	0.216126	0.108063	LC/MS/MS
Iprodione	36734-19-7	Dicarboximide	Fungicide	66617	2.4	1.2	3.8	1.9	66618	0.426668	0.213334	LC/MS/MS
Isofetamid	875915-78-9	Amide	Fungicide	53569	3.3	1.7	3	1.5	54386	0.20363	0.101815	LC/MS/MS
Kresoxim-methyl	143390-89-0	Strobin	Fungicide	67670	2.2	1.1	3.1	1.6	67672	0.260404	0.130202	LC/MS/MS
Malaoxon	1634-78-2	Organophosphorus	Insecticide degradate	68240	1.4	0.7	3.8	1.9	68242	0.210589	0.105295	LC/MS/MS
Malathion	121-75-5	Organophosphorus	Insecticide	65087	2.2	1.1	4	2	39531	0.219113	0.109557	LC/MS/MS
Mandestrobin	173662-97-0	Strobin	Fungicide	54358	3.2	1.6	3.3	1.7	54439	0.199938	0.099969	LC/MS/MS
Mandipropamid	374726-62-2	Amide	Fungicide	51854	2.6	1.3	4.6	2.3	54387	0.232985	0.116493	LC/MS/MS
Metalaxyl	57837-19-1	Xylylalanine	Fungicide	68437	1.1	0.6	4.4	2.2	68439	0.194749	0.097375	LC/MS/MS
Metalaxyl Alanine	85933-49-9	Xylylalanine	Fungicide degradate	54345	2.5	1.3	4	2	54416	0.221448	0.110724	LC/MS/MS
Metconazole	125116-23-6	Azole	Fungicide	66620	2.1	1	4.1	2.1	66622	0.207029	0.103515	LC/MS/MS
Methoprene	40596-69-8	Juvenile hormone mimic	Insect growth regulator	66623	11.5	5.8	13.5	6.8	NA	NA	NA	GC/MS/MS
Methoxyfenozide	161050-58-4	Diacylhydrazine	Insecticide	68647	1.9	1	3.1	1.5	54388	0.205869	0.102935	LC/MS/MS
Methylparathion	298-00-0	Organophosphorus	Insecticide	NA	NA	NA	NA	NA	39601	0.4	0.2	LC/MS/MS
Metolachlor	51218-45-2	Chloroacetanilide	Herbicide	65090	3.1	1.5	3	1.5	4005	0.2072	0.1036	LC/MS/MS
Myclobutanil	88671-89-0	Azole	Fungicide	66632	1.1	0.6	4.2	2.1	66634	0.196052	0.098026	LC/MS/MS
Naled (Dibrom)	300-76-5	Organophosphorus	Insecticide	68654	21.1	10.6	23.7	11.8	NA	NA	NA	LC/MS/MS
Napropamide	15299-99-7	Amide	Herbicide	65092	2	1	3	1.5	64164	0.204448	0.102224	LC/MS/MS
Nitrapyrin	1929-82-4	Chloropyridine	Nitrogen stabilizer	52763	2.1	1.1	3.3	1.6	54448	0.190152	0.095076	GC/MS/MS
Novaluron	116714-46-6	Benzoylurea	Herbicide	68655	4.5	2.2	4.4	2.2	NA	NA	NA	LC/MS/MS
Oryzalin	19044-88-3	2,6-Dinitroaniline	Herbicide	68663	3.8	1.9	3.2	1.6	54389	0.834211	0.417106	LC/MS/MS
Oxadiazon	19666-30-9	Unclassified	Herbicide	51843	1.7	0.9	3.9	1.9	68903	0.424108	0.212054	LC/MS/MS
Oxathiapiprolin	1003318-67-9	Pyrazole	Fungicide	52766	2.7	1.4	3	1.5	54390	0.217217	0.108608	LC/MS/MS

Table 3.1. Method detection and reporting limits for pesticides dissolved in water and sediments measured by the U.S. Geological Survey Organic Chemistry Research Laboratory.—Continued

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Compound	CAS number	Chemical class	Pesticide type	NWIS water and suspended sediment parameter code	Water RL (ng/L)	Water MDL (ng/L)	Suspended sediment RL (ng/L)	Suspended sediment MDL (ng/L)	NWIS sediment parameter code	Bed sediment RL (ng/g)	Bed sediment MDL (ng/g)	Analytical instrument
Oxyfluorfen	42874-03-3	Diphenyl ether	Herbicide	65093	2.7	1.4	2.5	1.3	64165	0.383934	0.191967	LC/MS/MS
p,p'-DDD	72-54-8	Organochlorine	Insecticide degradate	65094	2.7	1.3	2.3	1.1	39311	0.2	0.1	GC/MS/MS
p,p'-DDE	72-55-9	Organochlorine	Insecticide degradate	65095	3	1.5	2.5	1.2	39321	0.208799	0.1044	GC/MS/MS
p,p'-DDT	50-29-3	Organochlorine	Insecticide	65096	2.7	1.3	3.6	1.8	39301	0.196335	0.098168	GC/MS/MS
Paclobutrazol	76738-62-0	Azole	Plant growth regulator	51846	2.2	1.1	4.5	2.3	51867	0.401709	0.200854	LC/MS/MS
Pendimethalin	40487-42-1	2,6-Dinitroaniline	Herbicide	65098	2	1	3.9	2	64167	0.389137	0.194568	LC/MS/MS
Penoxsulam	219714-96-2	Triazolopyrimidine	Herbicide	51863	4.4	2.2	NA	NA	54391	0.489486	0.244743	LC/MS/MS
Pentachloroanisole	1825-21-4	Organochlorine	Insecticide degradate	66637	2.3	1.1	4.7	2.3	49460	0.420845	0.210423	GC/MS/MS
Pentachloronitrobenzene	82-68-8	Substituted Benzene	Fungicide	66639	2.9	1.4	6	3	49446	0.400941	0.200471	GC/MS/MS
Penthiopyrad	183675-82-3	Pyrazole	Fungicide	52769	2.2	1.1	3.9	1.9	54392	0.199366	0.099683	LC/MS/MS
Permethrin	52645-53-1	Pyrethroid	Insecticide	65099	1.4	0.7	1.5	0.7	64168	0.396211	0.198106	GC/MS/MS
Phenothrin	26002-80-2	Pyrethroid	Insecticide	65100	2.2	1.1	2.6	1.3	65112	0.912072	0.456036	GC/MS/MS
Phosmet	732-11-6	Organophosphorus	Insecticide	65101	1.4	0.7	3.3	1.6	64169	0.239977	0.119988	LC/MS/MS
Picarbutrazox	500207-04-5	Pyridine	Fungicide	54357	2.7	1.3	3.2	1.6	54438	0.205177	0.102589	LC/MS/MS
Picoxystrobin	117428-22-5	Strobin	Fungicide	51850	2.6	1.3	4.1	2	51871	0.199087	0.099544	LC/MS/MS
Piperonyl butoxide	51-03-6	Unclassified	Synergist	65102	2.1	1	4.3	2.1	64170	0.247365	0.123683	LC/MS/MS
Prodiamine	29091-21-2	2,6-Dinitroaniline	Herbicide	51844	2.2	1.1	4.1	2.1	68905	0.892833	0.446416	LC/MS/MS
Prometon	1610-18-0	Triazine	Herbicide	67702	2.9	1.4	2.8	1.4	82402	0.209511	0.104755	LC/MS/MS
Prometryn	7287-19-6	Triazine	Herbicide	65103	1.4	0.7	3.3	1.7	78688	0.224362	0.112181	LC/MS/MS
Propanil	709-98-8	Anilide	Herbicide	66641	2.5	1.2	3.8	1.9	66642	0.420223	0.210111	LC/MS/MS
Propargite	2312-35-8	Unclassified	Insecticide	68677	2.4	1.2	3.4	1.7	NA	NA	NA	LC/MS/MS
Propiconazole	60207-90-1	Azole	Fungicide	66643	1.5	0.7	2.6	1.3	66645	0.220249	0.110125	LC/MS/MS
Propyzamide	23950-58-5	Amide	Herbicide	67706	2.1	1	3.7	1.9	67708	0.402295	0.201147	LC/MS/MS
Pydiflumetofen	1228284-64-7	Pyrazole	Fungicide	54359	2.1	1	4.1	2	54440	0.248793	0.124397	LC/MS/MS
Pyraclostrobin	175013-18-0	Strobin	Fungicide	66646	2.9	1.5	3.6	1.8	66648	0.233312	0.116656	LC/MS/MS
Pyridaben	96489-71-3	Pyridazinone	Insecticide	68682	2.7	1.4	2.6	1.3	68909	0.240657	0.120329	LC/MS/MS
Pyrimethanil	53112-28-0	Pyrimidine	Fungicide	67717	2.6	1.3	2.2	1.1	NA	NA	NA	LC/MS/MS

Table 3.1. Method detection and reporting limits for pesticides dissolved in water and sediments measured by the U.S. Geological Survey Organic Chemistry Research Laboratory.—Continued

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Compound	CAS number	Chemical class	Pesticide type	NWIS water and suspended sediment parameter code	Water RL (ng/L)	Water MDL (ng/L)	Suspended sediment RL (ng/L)	Suspended sediment MDL (ng/L)	NWIS sediment parameter code	Bed sediment RL (ng/g)	Bed sediment MDL (ng/g)	Analytical instrument
Pyriproxyfen	95737-68-1	Hormone mimic	Insecticide	68683	2.3	1.1	3.3	1.7	54393	0.243426	0.121713	LC/MS/MS
Quinoxyfen	124495-18-7	Quinoline	Fungicide	51847	2.3	1.1	3.4	1.7	51868	0.208236	0.104118	LC/MS/MS
Resmethrin	10453-86-8	Pyrethroid	Insecticide	NA	NA	NA	NA	NA	65113	0.88	0.44	GC/MS/MS
Sedaxane	874967-67-6	Anilide, Pyrazole	Fungicide	52648	1.8	0.9	3	1.5	54394	0.220046	0.110023	LC/MS/MS
Simazine	122-34-9	Triazine	Herbicide	65105	1.7	0.9	2.7	1.4	39046	0.414632	0.207316	LC/MS/MS
Sulfometuron-Methyl	74222-97-2	Sulfonyl urea	Herbicide	NA	NA	NA	NA	NA	54420	0.46	0.23	LC/MS/MS
Sulfoxaflor	946578-00-3	Sulfoximine	Insecticide	52767	2.4	1.2	4.8	2.4	54395	0.412498	0.206249	LC/MS/MS
tau-Fluvalinate	102851-06-9	Pyrethroid	Insecticide	65106	1.6	0.8	2.1	1.1	65114	0.197193	0.098596	GC/MS/MS
Tebuconazole	107534-96-3	Azole	Fungicide	66649	1.3	0.6	4.6	2.3	66650	0.223265	0.111633	LC/MS/MS
Tebuconazole t-Butylhydroxy	212267-64-6	Azole	Fungicide degradate	54348	1.3	0.7	NA	NA	NA	NA	NA	LC/MS/MS
Tebufenozide	112410-23-8	Moulting hormone agonist	Insecticide	68692	2.4	1.2	3	1.5	54396	0.200387	0.100193	LC/MS/MS
Tebupirimfos	96182-53-5	Organophosphorus	Insecticide	68693	2.5	1.3	4.6	2.3	68913	0.209465	0.104733	LC/MS/MS
Tebupirimfos oxon	1035330-36-9	Organophosphorus	Insecticide degradate	68694	1.5	0.8	2.8	1.4	68911	0.212762	0.106381	LC/MS/MS
Tefluthrin	79538-32-2	Pyrethroid	Insecticide	67731	1.3	0.7	2.4	1.2	67733	0.196139	0.098069	GC/MS/MS
Tetraconazole	112281-77-3	Azole	Fungicide	66654	1.2	0.6	4.6	2.3	66656	0.223293	0.111647	LC/MS/MS
Tetramethrin	7696-12-0	Pyrethroid	Insecticide	66657	1.9	0.9	2.7	1.4	66659	0.38649	0.193245	LC/MS/MS
Thiabendazole	148-79-8	Benzimidazole	Fungicide	67161	3.4	1.7	4.5	2.2	NA	NA	NA	LC/MS/MS
Thiacloprid	111988-49-9	Neonicotinoid	Insecticide	68485	2.5	1.2	4.3	2.2	54398	0.227131	0.113566	LC/MS/MS
Thiamethoxam	153719-23-4	Neonicotinoid	Insecticide	68245	1.1	0.5	3.5	1.7	68247	0.206244	0.103122	LC/MS/MS
Thiamethoxam Degradate (NOA-355190)	902493-06-5	Neonicotinoid	Insecticide degradate	53568	2.9	1.4	5.2	2.6	54399	0.20702	0.10351	LC/MS/MS
Thiamethoxam Degradate (NOA-407475)	Not Available	Neonicotinoid	Insecticide degradate	53576	5.4	2.7	NA	NA	NA	NA	NA	LC/MS/MS
Thiobencarb	28249-77-6	Thiocarbamate	Herbicide	65107	2.4	1.2	4	2	64171	0.215102	0.107551	LC/MS/MS
Tolfenpyrad	129558-76-5	Pyrazole	Insecticide	51866	3.3	1.6	3.5	1.7	54400	0.252219	0.126109	LC/MS/MS
Triadimefon	43121-43-3	Azole	Fungicide	67741	1.5	0.8	3.4	1.7	67743	0.392488	0.196244	LC/MS/MS
Triadimenol	55219-65-3	Azole	Fungicide	67746	2.4	1.2	2.2	1.1	67748	0.472943	0.236472	LC/MS/MS
Triallate	2303-17-5	Thiocarbamate	Herbicide	68710	9.4	4.7	9.6	4.8	68919	0.818329	0.409164	LC/MS/MS

Table 3.1. Method detection and reporting limits for pesticides dissolved in water and sediments measured by the U.S. Geological Survey Organic Chemistry Research Laboratory.—Continued

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Compound	CAS number	Chemical class	Pesticide type	NWIS water and suspended sediment parameter code	Water RL (ng/L)	Water MDL (ng/L)	Suspended sediment RL (ng/L)	Suspended sediment MDL (ng/L)	NWIS sediment parameter code	Bed sediment RL (ng/g)	Bed sediment MDL (ng/g)	Analytical instrument
Tribufos	78-48-8	Organophosphorus	Defoliant	68711	2.8	1.4	2.2	1.1	39050	0.236903	0.118451	LC/MS/MS
Trifloxystrobin	141517-21-7	Strobin	Fungicide	66660	2.6	1.3	4	2	66662	0.204114	0.102057	LC/MS/MS
Triflumizole	68694-11-1	Azole	Fungicide	67753	2.5	1.3	3.1	1.6	67755	0.194873	0.097437	LC/MS/MS
Trifluralin	1582-09-8	2,6-Dinitroaniline	Herbicide	65108	2.6	1.3	4.3	2.2	62902	0.186783	0.093391	GC/MS/MS
Triticonazole	131983-72-7	Azole	Fungicide	67758	2.6	1.3	3.7	1.9	67760	0.199992	0.099996	LC/MS/MS
Valifenalate	283159-90-0	Acylamino Acid	Fungicide	54360	2	1	4.8	2.4	54441	0.179406	0.089703	LC/MS/MS
Vinclozolin	50471-44-8	Dicarboximide	Fungicide	NA	NA	NA	NA	NA	67765	0.179406	0.089703	GC/MS/MS
Zoxamide	156052-68-5	Amide	Fungicide	67768	1.7	0.8	3.8	1.9	67770	0.3946	0.1973	LC/MS/MS

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